

RAUI: Uncertainty Indicators Built With Artificial Intelligence

Morteza Ghomi*

Banco de España

Samuel Hurtado[†]

Banco de España

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Abstract

We present a novel methodology to generate uncertainty indicators for user-defined topics using newspaper text data. The approach builds on Retrieval-Augmented Generation (RAG) systems commonly used in artificial intelligence applications, which we adapt to construct Retrieval-Augmented Uncertainty Indicators (RAUI). The method employs semantic search with an embedding model to select news articles relevant to a given topic, and a large language model (LLM) to quantify the level of uncertainty conveyed in those articles. Using Spanish newspaper data, we construct uncertainty indicators for ten topics and an aggregate measure that also highlights how each topic contributes to overall uncertainty. We present practical applications of these indicators and conduct a VAR analysis to examine how different sources of uncertainty affect the Spanish economy.

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*Banco de España, Madrid (Spain) morteza.ghomi@bde.es.

[†]Banco de España, Madrid (Spain) samuel.hurtado@bde.es.

1 Introduction

Text-based indicators of sentiment and uncertainty are widely used in economics because they provide timely and interpretable measures of expectations, policy perceptions, and confidence. Most existing indicators rely on dictionary methods, which typically proceed in two steps: first, relevant articles are selected using topic-specific keywords; second, the uncertainty or sentiment content of those texts is quantified using predefined word lists associated with ambiguity, negativity, or risk. This approach is transparent and easy to implement, but it has important limitations. Its performance is highly sensitive to the exact wording of articles, it struggles to capture context-dependent meanings, and it often weakens as language evolves or terms acquire new connotations. More fundamentally, dictionary methods treat words largely in isolation, abstracting from the semantic and syntactic structure that shapes meaning at the sentence or paragraph level. As a result, they can misclassify both relevance and tone, leaving ample scope to improve text-based measures of uncertainty.

This paper proposes a new approach that uses state-of-the-art artificial intelligence techniques to improve both stages of text-based indicator construction. We develop *Retrieval-Augmented Uncertainty Indicators* (RAUI), a methodology that combines semantic retrieval with contextual uncertainty scoring. This approach is inspired by Retrieval-Augmented Generation (RAG) architectures commonly used in AI-based chatbots. In RAG systems, relevant documents are first identified through a semantic search using text embeddings, and only a subset of those documents is passed to an LLM for reasoning. By adapting this framework to the construction of uncertainty indicators, we can harness the interpretative capabilities of modern language models while maintaining computational feasibility.

Following RAG systems, our methodology exploits AI in both stages of constructing indicators. In the first stage, we use a text-embedding model to perform semantic search over a large corpus of newspaper articles and retrieve those that are most closely related to a user-defined topic. This replaces traditional keyword filtering with a retrieval procedure based on similarity in meaning, allowing the method to identify conceptually related articles even when they do not contain the exact terms included in a dictionary. To select an appropriate embedding model tailored to our specific use case, we conducted an empirical evaluation and compared the performance of different models to human classification of a random sample of articles.

In the second stage, we pass the retrieved articles to a large language model, which evalu-

ates the degree of uncertainty expressed in each article with respect to the relevant topic. This replaces word counting with a contextual evaluation of the text as a whole, making it possible to distinguish between articles that merely mention a topic and articles that genuinely express uncertainty about it. This includes providing detailed prompts with several topic-related examples to guarantee the consistency of uncertainty quantification across articles. The core idea behind the RAUI method is therefore to use modern AI tools to improve both *selection* and *quantification*, while preserving tractability by applying the language model only to the subset of texts identified as relevant by the retrieval stage.

The RAUI framework improves on traditional text-based approaches along the two margins that matter most. First, it improves retrieval by moving from keyword matching to semantic search in embedding space, thereby reducing false classifications. Second, it improves measurement by replacing counts of uncertainty-related words with topic-specific evaluations that interpret articles in context. Because only the retrieved subset of articles is passed to an LLM, the method remains computationally feasible even for databases containing millions of news items. RAUI adapts the logic of retrieval-augmented generation systems to the construction of economic indicators, allowing us to exploit the interpretative capabilities of LLMs without requiring full-scale LLM processing of the entire corpus.

We apply this framework to more than four million Spanish news articles published between 2000 and 2025 and construct ten topic-specific uncertainty indicators: geopolitical tensions; international trade and tariffs; export markets and external demand; financial markets; energy and raw-material prices; inflation, prices, and markups; wages and wage bargaining; fiscal policy and the sustainability of public debt; housing; and confidence and internal demand. After evaluating several embedding models on a manually labeled sample of articles, we select the *mle5l* model, a multilingual embedding architecture that performs best at retrieving Spanish-language articles associated with economically meaningful topics. We then use GPT-4o-mini to score the uncertainty content of roughly 130,000 articles per topic.

A key feature of the framework is that each topic-level indicator combines two components. The first is a *signal*, defined as the average level of uncertainty expressed in the articles associated with that topic. The second is an *amplifier*, defined as the relative prominence of the topic in the news. This distinction is economically meaningful. A topic may be discussed in highly uncertain terms but receive little attention, or it may dominate public discourse even when the uncertainty expressed in each article is only moderate. The topic-specific RAUI is defined as

the interaction of these two components, and the aggregate RAUI is a weighted combination of topic-level signals where the weights are given by topic prevalence. This structure delivers not only an overall measure of uncertainty, but also a decomposition that shows how each topic contributes to aggregate uncertainty over time.

The resulting indicators reveal that aggregate uncertainty masks substantial heterogeneity in composition. Aggregate uncertainty in Spain rises sharply during well-known episodes such as the early-2000s, the global financial and sovereign debt crises, the COVID-19 pandemic, the Russian invasion of Ukraine, and the tariff-related tensions of 2025. But the underlying sources of uncertainty differ markedly across episodes. During 2002–2003, uncertainty is driven mainly by external factors, including geopolitics, trade, and global demand. During 2008–2013, by contrast, uncertainty is driven predominantly by domestic factors, including housing, financial markets, prices, fiscal policy, and wages. With the arrival of COVID-19, nearly all topics rise simultaneously. In 2022, geopolitical tensions, international trade, inflation, and energy price uncertainty pushed the aggregate RAUI to exceptionally high levels. These patterns are difficult to recover from aggregate uncertainty indicators alone. By construction, RAUI allows us to move from a single scalar index to a richer representation of the origins of uncertainty and their changing importance over time.

Comparing RAUI with existing uncertainty measures also helps clarify what those indicators capture. We show that dictionary-based indicators, such as the Geopolitical Risk Index (GPR), tend to comove more strongly with the prominence of topic-specific news coverage, that is the amplifier component of RAUI. By contrast, market-based indicators, such as financial market volatility indices, comove more closely with the signal component, which captures the intensity of uncertainty expressed in the text. In this sense, RAUI provides a useful bridge across existing measures, showing that different indicators place greater weight on distinct margins of the uncertainty process.

To illustrate the economic relevance of these measures, we use them in a monthly VAR for the Spanish economy spanning 2000M1 to 2025M12. A shock to the aggregate uncertainty index generates a persistent decline in industrial production, an increase in unemployment, a fall in stock prices, and a monetary-policy response consistent with accommodation. These results are broadly in line with the existing literature. However, the richer contribution of our framework comes from replacing the aggregate measure with topic-specific indicators. Doing so reveals substantial heterogeneity in the transmission of uncertainty shocks. Uncertainty re-

lated to inflation and markups, fiscal policy, housing, internal demand, and financial markets produces larger and more persistent macroeconomic effects than uncertainty related to more external topics such as geopolitical tensions or international trade. These findings suggest that uncertainty should not be treated as a homogeneous disturbance. Aggregate uncertainty reflects a combination of conceptually distinct forces, and their economic importance depends crucially on whether they originate in domestic macro-financial conditions or in external developments.

The paper makes three main contributions. First, it introduces a more advanced methodology for constructing uncertainty indicators from text by combining semantic retrieval with contextual quantification. Relative to standard text-based approaches, the method is more robust to changes in language, less dependent on predefined word lists, and computationally efficient. Second, the RAUI methodology offers a flexible and agile way to construct topic-specific indicators and provides a visualization that shows how each topic contributes to the aggregate uncertainty measure. Third, by applying the methodology to the Spanish economy, the paper shows empirically that the composition of uncertainty matters for macroeconomic analysis: different sources of uncertainty contribute differently to aggregate fluctuations and generate distinct dynamic responses in the economy.

The remainder of the paper proceeds as follows. After briefly reviewing the relevant literature on uncertainty measurement, Section 2 presents the RAUI methodology, including the retrieval and quantification stages and the validation exercises used to select the embedding model. Section 3 describes the resulting topic-specific and aggregate indicators for Spain. Section 4 studies their macroeconomic effects using VAR analysis and Section 5 concludes.

Related Literature

The literature has developed several approaches to measuring economic uncertainty. These range from using financial-market volatility as a forward-looking proxy for uncertainty (Bloom, 2009; Baker et al., 2020), to inferring uncertainty from disagreement in the expectations of firms, households, or professional forecasters (Bachmann et al., 2013; Jo and Sekkel, 2019; Gambetti et al., 2025), to model-based approaches that construct uncertainty from the unpredictable component of macroeconomic variables (Jurado et al., 2015; Rossi and Sekhposyan, 2015; Comunale and Nguyen, 2023). These measures have been highly influential, but they are less suited to capturing uncertainty at high frequency and in a topic-specific way. This limitation has con-

tributed to the growing use of text-based indicators.

In recent decades, text-based indices derived from news have become central tools for measuring economic uncertainty. The seminal example is the Economic Policy Uncertainty (EPU) index developed by [Baker et al. \(2016\)](#). This index is constructed by counting newspaper articles that contain a trio of keywords related to “economy,” “policy,” and “uncertainty,” using Boolean logic to identify relevant articles. The index has been computed for many countries (see, e.g., [Ghirelli et al., 2019](#) for Spain) and captures spikes corresponding to periods of heightened uncertainty. [Baker et al. \(2022\)](#) also develop a state-level measure of EPU using local newspapers in the United States. Beyond the generic EPU, similar keyword-count indices have been developed for specific domains. For example, [Caldara and Iacoviello \(2022\)](#) propose a Geopolitical Risk (GPR) index based on news about international tensions, [Caldara et al. \(2020\)](#) introduce a Trade Policy Uncertainty index using trade-related terms, [Ahir et al. \(2022\)](#) construct the World Uncertainty Index (WUI) from Economist Intelligence Unit reports across 143 countries, and [Baker et al. \(2021\)](#) build a Twitter-based measure from tweets mentioning “economy” and “uncertainty.” News-based indices are forward-looking, capturing uncertainty as perceived in real time, but early methods relied on fixed keywords and simple word counts. This limited their ability to capture subtleties in language and required manual adjustments to incorporate new sources of uncertainty, such as novel terms like “Brexit” or “Covid-19.”

Some recent contributions employ artificial intelligence and advanced natural language processing to construct uncertainty indicators that are both more flexible and more granular, thereby mitigating the reliance on fixed dictionaries. A prominent example is the use of unsupervised machine learning techniques to identify latent topics in news text and to measure uncertainty at the topic level. [Azqueta-Gavaldón \(2017\)](#) first applied Latent Dirichlet Allocation (LDA) to newspapers to automatically extract themes from a large collection of news articles¹, while [Larsen \(2021\)](#) uses LDA to disentangle different types of uncertainty. [Rauh \(2019\)](#) applies it to develop regional EPU indices for Canada, and [Yono et al. \(2020\)](#) propose a supervised LDA to ensure that the resulting topics are directly related to financial market volatility. Particularly closely related contributions are [Audrino et al. \(2026\)](#) and [Iacoviello and Tong \(2026\)](#), who construct uncertainty and risk measures by directly applying GPT-4 models to newspaper articles and show that the resulting indicators outperform traditional keyword-based methods. While their framework relies on direct LLM classification of news articles, our

¹[Azqueta-Gavaldon et al. \(2020\)](#) use this method to construct an EPU index for selected euro area countries.

RAUI methodology combines semantic retrieval using embeddings with LLM-based quantification, making it computationally scalable and naturally suited for constructing topic-specific uncertainty indicators and uncertainty decompositions.

Another AI-oriented advancement uses distributed word embeddings to expand and refine the set of uncertainty-related terms. Word embedding models learn vector representations of words from large text corpora, capturing semantic similarities. Researchers have applied this approach to improve keyword-based indices (Tsai and Wang, 2014; Theil et al., 2020; Kaveh-Yazdy and Zarifzadeh, 2021; Naboka-Krell, 2024), and empirical tests show that using expanded keyword sets often improves performance (Keith et al., 2020). A more direct alternative is to move beyond keyword expansion altogether and train text classifiers to detect economic uncertainty. This requires labeled datasets of relevant and non-relevant articles for model training or fine-tuning. For instance, Keith et al. (2020) used 2,500 hand-labeled articles from the Baker et al. (2016) corpus to train a bag-of-words classifier. More recently, transformer-based language models have been used to better capture context and semantics. Qureshi et al. (2022) constructed EPU indices for the U.S. and Canada using multiple state-of-the-art NLP tools, while Yeh et al. (2024) piloted GPT-based keyword generation for the EPU index, showing that the derived measures are robust.

Our paper advances the uncertainty-measurement literature by introducing a Retrieval-Augmented Generation framework that constructs topic-specific and aggregate uncertainty indices from newspaper text. Unlike traditional keyword or dictionary approaches, RAUI combines semantic retrieval using text embeddings with quantification of uncertainty using a large language model, enabling context-aware classification and reducing errors arising from polysemous terms, irrelevant matches, and linguistic complexity. Because only retrieved passages are processed by the LLM, the method remains computationally efficient and scalable for large corpora.

2 Methodology

2.1 RAG

In recent years, Retrieval-Augmented Generation (RAG) has emerged as a foundational technique in artificial intelligence, particularly in natural language processing applications that require grounding large language models (LLMs) in specific knowledge sources. The approach,

first formalized by [Lewis et al. \(2020\)](#), integrates two distinct but complementary components: an information retrieval system and a generative model. The retrieval component searches a knowledge database and identifies the set of documents or text fragments that are semantically closest to a given query, while the generation component (typically a pretrained transformer-based LLM) produces a natural language response conditioned on the retrieved material. This dual structure enables systems to deliver outputs that are both linguistically coherent and empirically anchored in a specific corpus of information.

The rationale for RAG systems arises from a key limitation of contemporary LLMs: despite their impressive capacity for general linguistic competence, they are constrained by the static and generic nature of their training data. As a result, their internalized knowledge becomes outdated over time and cannot easily incorporate new or proprietary information without costly retraining. RAG systems address this constraint by decoupling knowledge storage from language and reasoning. Instead of encoding all domain knowledge within the model weights, RAG architectures retrieve relevant data from an external database at inference time, providing the model with the necessary contextual evidence to formulate an informed response. This modularity allows continuous updating and specialization of the knowledge base without retraining the model.

In practical terms, RAG methodologies have found broad application in cases where accuracy, transparency, and domain specificity are crucial. A typical example arises in the legal or regulatory sphere, where a RAG-based system might allow users to query an extensive database of judicial decisions. Upon receiving a natural-language question (e.g. a user asking about the prevailing direction of court rulings on a specific legal argument), the retrieval component searches the corpus of past decisions, selecting the most semantically relevant excerpts. These passages are then presented to an LLM as contextual evidence, enabling the model to read them and generate an answer explicitly grounded in the retrieved texts. The resulting response is therefore both contextually tailored and verifiable, since the retrieved materials serve as citations or evidence underlying the model’s output. Also, because the LLM is constrained to base its response on the retrieved content, its output tends to be less prone to hallucinations, more verifiable and less speculative (e.g., [Shuster et al., 2021](#); [Izacard et al., 2023](#)). For these reasons, the methodology has rapidly become a standard tool for constructing specialized assistants, question-answering systems, and domain-adapted text analysis pipelines.

Building on this framework, we propose a Retrieval-Augmented Uncertainty Indicator

(RAUI), which adapts the logic of RAG systems to the construction of economic uncertainty measures derived from textual data. In our implementation, the retrieval component operates over a large-scale corpus of newspaper articles, identifying those items that belong to a user-defined topic, such as geopolitics, international trade, inflation, wages, etc. The selected texts for each topic are then passed to an LLM for analysis and scoring, producing a structured representation of the level of uncertainty that was measured in the discourse within each topic. Through this adaptation, the RAG paradigm serves not as a conversational interface, but as a methodological foundation for the quantitative modeling of uncertainty based on unstructured textual evidence.

The remaining subsections in this section describe, respectively, the mechanisms through which relevant items are identified, and the architecture of the interaction layer that governs the LLM’s processing of this contextual information.

2.2 Semantic Search Using an Embedding Model

The retrieval component in Retrieval-Augmented Generation (RAG) architectures typically relies on semantic search, a methodology that evaluates the similarity between textual elements not by surface-level word matching, but by their meaning as represented in a continuous vector space. This process is made possible through the use of text embedding models, which transform natural language into numerical representations that capture semantic and syntactic relationships between words, phrases, or entire documents.

A text embedding model can be understood as a function that projects textual input into a high-dimensional vector space, often with thousands of dimensions (e.g., 1024, 2048, or 4096). Within this space, semantically similar texts are mapped to points that are close to one another. Well-constructed embeddings thus encode aspects of linguistic meaning in geometric form. A classic example illustrates this property through analogical relationships: the vector representations of king, queen, prince, and princess are usually situated in proximity, and the difference between queen and king vectors is similar to the one between princess and prince, that is, $\text{embedding}(\text{prince}) + [\text{embedding}(\text{queen}) - \text{embedding}(\text{king})] \approx \text{embedding}(\text{princess})$. This capacity to capture analogical reasoning and conceptual similarity underpins most modern approaches to semantic retrieval.

Unlike earlier models that relied on word co-occurrence statistics or bag-of-words representations, contemporary embedding models are based on transformer architectures ([Vaswani](#)

et al., 2017). These architectures process text in context, meaning that a word’s representation dynamically depends on the surrounding words in the sentence or paragraph. Consequently, the same term may be mapped to distinct regions of the embedding space depending on context. For instance, the phrase white king is likely to appear in a discussion of chess and is therefore expected to cluster near vectors such as rook or pawn, rather than near queen or royalty. This contextual sensitivity allows embedding models to capture nuanced distinctions in meaning that are essential for accurate semantic search.

Embedding models form an integral component of LLMs, serving as the initial stage of the language-processing pipeline. However, specialized models have also been developed exclusively to generate high-quality embeddings. Such standalone embedding models, including families such as OpenAI’s text-embedding series, E5 (Wang et al., 2022), and Instructor (Su et al., 2023), are optimized specifically for measuring semantic similarity across texts. Their training objectives typically involve contrastive learning, whereby the model learns to assign higher similarity scores to semantically related text pairs and lower scores to unrelated ones. These models have become central tools in RAG systems, as the precision of their semantic retrieval directly determines the relevance and quality of the context supplied to the generative model.

In the context of our Retrieval-Augmented Uncertainty Indicators (RAUI), the choice of embedding model plays a critical role in determining which news items are deemed relevant for subsequent analysis by the LLM. In our framework, the embedding model is used in two stages: an initial pre-processing phase, in which the entire news database is projected into the embedding space, and a subsequent retrieval phase, in which each topic query is similarly projected and compared with the pre-computed vectors of all news items. During the pre-processing stage, every news piece in the corpus is transformed into its corresponding vector representation. This operation is computationally intensive, but it only needs to be performed once, since the resulting vectors can be stored and reused for future searches. Although this stage requires significant processing time, it remains substantially faster than passing the entire corpus through a generative model². The retrieval stage involves projecting each thematic query (in our case, the title of each topic, e.g. “inflation and prices” or “international trade”) into the same semantic space. We then compute the cosine similarity between the query vector

²Our selected model is approximately 200 times faster than using deepseek-7B locally in Ollama to process the full database on the same computer, and approximately 20 times faster than paying to do so with GPT-4o-mini in the cloud.

and each pre-computed news vector, and use that to rank articles from most to least similar. This method ensures that conceptually similar articles, regardless of their specific wording, are retrieved with high priority.

Selecting an appropriate embedding model is nontrivial, as performance depends on language, domain, and contextual nuances. Although public benchmarks evaluate models on standard semantic similarity tasks, these results do not always generalize to domain-specific applications. A model that performs well in English, for example, may perform poorly in Spanish, and a model optimized for general discourse may fail to capture the terminology and structure of economic news. We therefore conducted an empirical evaluation tailored to our use case: identifying Spanish-language news articles related to a set of predefined topics. To construct a ground-truth dataset, we randomly sampled 443 articles from our database, corresponding approximately to one day of news coverage. A research assistant manually read each article and assigned binary relevance labels, allowing articles to belong to multiple topics. The classification covered 11 topics central to our framework: international trade, tourism, indicators of economic activity, inflation and prices, wages, employment, firms, debt, banks and financial institutions, financial markets, and energy.

Using this annotated dataset, we evaluated the retrieval performance of seven different embedding models, along with two alternative approaches: a dictionary-based method, corresponding to traditional keyword-matching procedures, and a direct LLM classification approach, in which GPT-4o-mini was asked to determine whether each news article belonged to each topic.³ For each model and each query topic, we computed the cosine similarity between the query vector and the vectors of all articles in the random sample, and ranked articles from most to least similar. We then compared these rankings with the manually labeled ground truth to assess how effectively each model retrieved relevant documents. The resulting performance graphs show, for each embedding model, the share of topic-relevant articles identified by the human coder (vertical axis) that appear within the first X ranked articles (horizontal axis). Figure 1 reports the results for the first topic, international trade.

To provide a concise quantitative summary of model performance across topics, we computed a Gini-like concentration index over the cumulative retrieval curves described above. Conceptually, this statistic measures the area between each model’s cumulative relevance curve

³Importantly, the OpenAI models were used in Azure AI Forge, a walled environment compatible with the terms and conditions under which Banco de España accesses its news database. The LLM-based classification approach is slower and more expensive than embedding-based retrieval, but remains feasible for small samples such as this evaluation exercise. GPT-4o-mini was likewise used within Azure AI Forge.

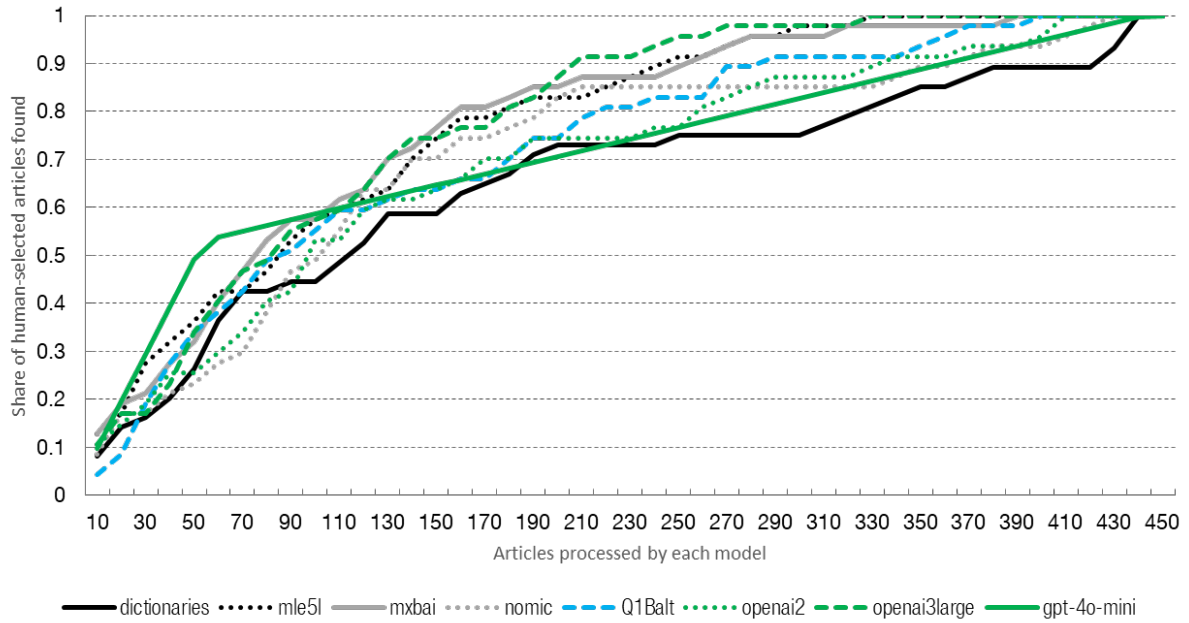


Figure 1: Accuracy of embedding models compared to human classification for the topic of international trade.

and the diagonal representing a uniform distribution of relevant articles along the ranked list.⁴ A higher value of this index indicates that relevant articles are concentrated toward the top of the ranking (that is, retrieved early) while irrelevant articles appear later. Conversely, a lower value suggests that relevant and irrelevant articles are more evenly distributed throughout the ranked list, reflecting a weaker discriminative power of the embedding model.

Intuitively, a well-performing model should retrieve most of the relevant news items within the first portion of the ranked list, producing a curve with a steep initial slope that quickly reaches a plateau at a level close to one. This corresponds to a high degree of “inequality” in the distribution of relevant hits across ranks, and therefore to a high Gini-like value. This Gini-like statistic thus offers a single, interpretable scalar measure that summarizes the retrieval efficiency of each embedding model, balancing ranking precision and recall concentration in a compact form.

To further summarize the results, we then average, for each model, the Gini-like indices across the eleven thematic queries used in our test exercise. Figure 2 reports this aggregated indicator that serves as a concise comparative summary of how effectively each embedding model distinguishes between relevant and irrelevant articles across a diverse set of economic

⁴While the same numerical formula as the traditional Gini index is applied, the ordering of observations differs: in our case, articles are sorted by similarity to the query rather than by magnitude of a variable. Consequently, the resulting measure is not identical to the conventional Gini coefficient and, in some cases, can be working with curves that fall below the diagonal.

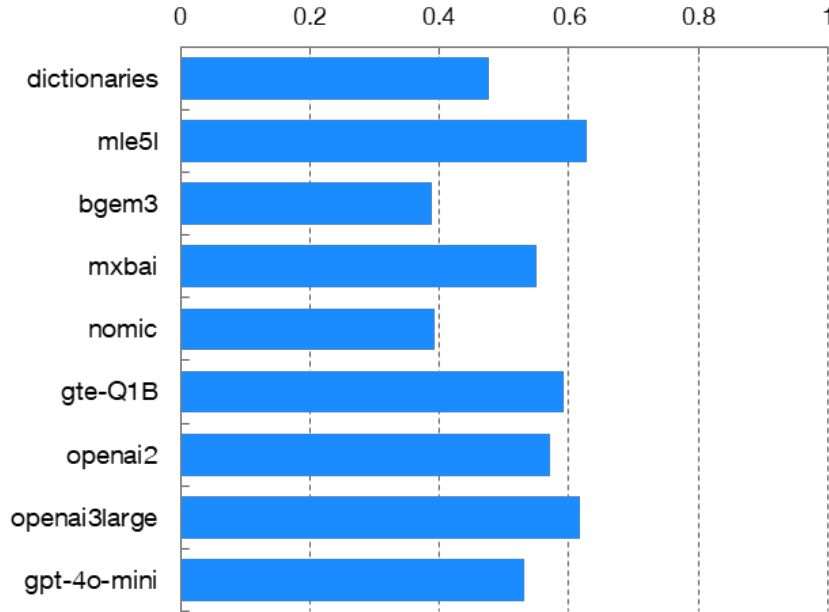


Figure 2: Gini-like index that measures the performance of each embedding model on average over the 11 topics.

topics. By condensing the performance of the models into a single quantitative measure, it allows for straightforward identification of the most promising candidates for large-scale deployment within the RAUI retrieval pipeline.

Based on the comparative evaluation described above, we selected the multilingual-e5-large (mle5l) model as the embedding model for the semantic retrieval stage in the construction of our uncertainty indicators. This model, developed by [Wang et al. \(2024\)](#) for Microsoft, belongs to the E5 family of multilingual embedding architectures and is publicly available through both Hugging Face and Ollama. Its accessibility through these platforms allows it to be executed either locally or within a private cloud environment, ensuring compliance with our data licensing requirements while remaining free to use. In practical terms, mle5l offers an advantageous balance between performance and computational efficiency: with approximately 560 million parameters, it is substantially smaller than typical large language models but still large enough to capture nuanced semantic relationships across multiple languages, including Spanish. Processing our entire corpus, comprising more than four million news articles spanning the period 2000 to 2025, required roughly nine days of computation time.

Figure 3 provides a detailed comparison of the performance of the selected embedding model across the 11 topics in the exercise relative to conventional dictionary-based methods. The embedding model improves retrieval performance in every topic, indicating that semantic search can deliver consistent and meaningful gains in both accuracy and overall effectiveness.



Figure 3: Gini-like index measuring the performance of the selected embedding model (multilingual-e5-large) and the traditional dictionary-based method across the topics in the validation exercise.

These results point to a clear methodological advantage of embedding-based approaches over traditional dictionary strategies, especially when identifying relevant content requires capturing subtle semantic relationships between terms. Figure 3 also shows that some topics are inherently difficult to define. We initially intended to include a topic on “firms” and another on “indicators of economic activity,” but the exercise revealed that all selection methods - dictionaries, embedding models, and GPT- performed relatively poorly on both. The difficulty with “firms” is straightforward: firms are central actors in a wide range of events and therefore appear almost uniformly throughout the newspaper database, making the topic too diffuse to isolate effectively. By contrast, “indicators of economic activity” are more naturally examined outside the context of news articles, as they consist of numerical measures routinely tracked by forecasters and analysts. We therefore excluded both from the final set of topics and instead introduced a topic on geopolitics, which had been absent from the initial analysis.

While embedding-based retrieval provides a robust first approximation of topical relevance, it can be improved through a secondary LLM-based validation stage. Since the LLM is already prompted to assess the uncertainty expressed in each retrieved article, it can also verify whether the article genuinely pertains to the intended topic. This two-stage approach—semantic retrieval followed by LLM-based validation—reduces residual false positives arising from imperfect embedding-based discrimination. Figure A2 in the Appendix reports the same

Gini-like index for all models after GPT-4o-mini is used to eliminate false positives. Comparing the result with Figure 2 shows that the gains are largest for models with lower initial retrieval precision, although reliability also improves for stronger models such as mle5l.

Before concluding this section, it is worth highlighting several practical lessons we derived from our evaluation of embedding models for the construction of RAUI. These insights reflect both quantitative findings and qualitative observations that emerged during model testing and experimentation:

- Specialized embedding models outperform embedding layers from general-purpose LLMs. Models explicitly designed and trained for semantic search consistently delivered superior results compared with the embedding layers extracted from general-purpose LLMs (which we also tested but did not present here). The latter, although powerful in generative tasks, are not optimized for semantic similarity computations and therefore tend to produce less coherent retrieval results in practice.
- Minimal text preprocessing yields better performance. Traditional text-mining approaches often rely on extensive preprocessing (removing punctuation, stop words, or special characters, etc.) to reduce noise. However, in the case of transformer-based embedding models, such preprocessing can degrade semantic integrity, as these models are trained to interpret linguistic structure in context. We found that using the raw, unaltered text produced more reliable and semantically consistent embeddings.
- Concise queries outperform elaborate descriptions. Short, well-focused topic prompts generally yielded more accurate retrievals than longer or more detailed formulations. For instance, for the topic “inflation and prices”, a succinct three-word query significantly outperformed a 133-word version elaborating on the same theme. This suggests that excessive detailing of the query can introduce irrelevant dimensions of meaning that obscure the central concept of interest.
- Embedding models can differentiate fine-grained thematic distinctions. Semantic search using embeddings proved capable of distinguishing between closely related economic topics. For example, the topics “wages” and “employment” produced distinct sets of articles, suggesting that the embedding space encodes sufficient nuance to separate related aspects of the labor market.

- False positives are interpretable and contextually related. Even when the retrieval system produced false positives, these typically corresponded to semantically adjacent concepts rather than random noise. For instance, within the “wages” topic, non-relevant results often involved articles about unemployment benefits, layoffs, working hours, or productivity. This interpretability enhances confidence in the overall retrieval framework.
- Some topics remain intrinsically ambiguous. Certain conceptual categories are more difficult to delimit semantically, leading to lower retrieval precision. In our experiments, the topic “firms” exhibited the weakest performance across all models, reflecting the broad and multifaceted nature of the term. This insight led us to refine and disaggregate some topics before generating the final uncertainty indicators, thereby improving thematic clarity and model alignment.

Taking these lessons into account, we implemented the semantic retrieval step of the RAUI pipeline using the mle5l embedding model and a slightly revised set of topics:

1. Geopolitical tensions
2. International trade and tariffs
3. Export markets and external demand
4. Financial markets
5. Energy prices and prices of other raw materials
6. Inflation, prices and markups
7. Wages and wage bargaining
8. Fiscal policy and sustainability of public debt
9. Housing market
10. Confidence of agents and evolution of internal demand

For each of the ten finalized topics, we selected approximately 130,000 news articles, corresponding on average to 12 articles per topic per week per newspaper over the 2000–2025 period. A global cosine similarity threshold was estimated across the entire corpus until 2024 to maintain consistent selection criteria over time and across newspapers. This design implies that the number of articles evaluated by the LLM naturally increases during periods when a given topic receives greater media attention, allowing the indicator to remain sensitive to fluctuations in the intensity of public discourse.

This initial set of ten monthly indicators can be readily expanded to include additional

topics as new developments warrant. Constructing a new indicator from scratch (including testing with a small random sample, defining the final query, and generating the complete historical series) requires approximately one day in total, of which around 20 hours correspond to unattended computation. Updating the full set of indicators on a weekly basis takes roughly 90 minutes, again mostly consisting of unattended processing time.

2.3 Quantification Using an LLM

Once the relevant news articles have been identified through the semantic retrieval process, the next step in the construction of the RAUI involves quantifying the degree of economic uncertainty expressed in each text. For this purpose, we employed GPT-4o-mini, a compact large language model capable of processing text with high contextual sensitivity and relatively low cost. This model achieves good results in standard benchmarks that assess text understanding, such as MMLU Pro. Each selected article is independently analyzed by the model, which assigns a numerical value reflecting its assessed level of uncertainty⁵. The prompt design is a critical component of the evaluation of uncertainty. Our final choice is the result of several experiments where we assessed how variations of the prompt generated different model responses. The text provided to the LLM is intentionally detailed, guiding the model to deliver consistent, interpretable, and reproducible outputs.

When an article is classified under several topics, the model is queried separately for each topic to ensure that the resulting uncertainty evaluations are topic-specific and not influenced by the coexistence of other thematic categories. For example, a news article may discuss both tariffs and energy prices, and may convey higher uncertainty when talking about one of them than about the other. Each prompt asks about the level of uncertainty that the article displays when talking about one specific topic.

A further key element of the prompt is the inclusion of illustrative examples. Each topic-specific prompt contains a minimum of ten headlines (some drawn from the real dataset and others manually crafted) to serve as reference standards for the uncertainty scale. Each headline is accompanied by the uncertainty value that we would assign to it, forming a kind of calibration set for the LLM. These examples were found to be essential for stabilizing the model’s be-

⁵It is worth noting that this same methodological framework could readily be adapted to produce related indicators. For example, by adjusting the LLM prompt to focus on sentiment rather than uncertainty, one could derive a Retrieval-Augmented Sentiment Indicator (RASI). In fact, the prompt formulation that we used requests both an uncertainty and a sentiment score, even though only the uncertainty dimension is presented here.

havior and ensuring comparability across queries. Indeed, initial experiments conducted without these examples produced very noisy and, in some topics, practically meaningless results, underscoring the importance of prompt-based contextual anchoring in this type of application.

Example of a prompt (translated to English):

This is a news article that talks about fiscal policy and the sustainability of public finances.

Give me a numerical assessment of the sentiment (0 if very negative, 5 if neutral, 10 if very positive) and the uncertainty (0 if there is little uncertainty, 10 if there is a lot of uncertainty) that reflects how this news talks about fiscal policy, sustainability of public finances, taxes, subsidies, pensions, public spending, fiscal rules, government budgets, etc.

(if the news does not talk about any of these topics, simply respond None, None).

Give me a simple answer. If the evaluation is that the value for sentiment is V_s and the value for uncertainty is V_u , give the answer in the following format:

Sentiment V_s , uncertainty V_u .

Here are some examples of hypothetical news headlines, and what their assessment might be:

Garriga (Vox) predicts that Sánchez's rearmament plan will entail a hidden tax increase (Sentiment 4, Uncertainty 6).

PSOE of Calahorra denounces a 211 percent increase in debt and warns of more taxes (Sentiment 2, Uncertainty 7).

PP demands Sánchez to lower taxes for farmers in the face of US tariff uncertainty (Sentiment 5, Uncertainty 7).

The city council approves a budget of 13.2 million that prioritizes social spending, education, and the local economy without raising taxes (Sentiment 8, Uncertainty 2).

Trump threatens Harvard with withdrawing tax-free status after freezing 2.2 billion for not yielding to his political demands (Sentiment 3, Uncertainty 8).

PP government does not rule out further tax cuts in Cantabria: we may have some margin (Sentiment 7, Uncertainty 6).

Prime Minister of Portugal presents a program with lower taxes and more public housing (Sentiment 8, Uncertainty 2).

The State grants Navarra more tax collection powers for new taxes (Sentiment 5, Uncertainty 5).

The Senate urges the Government to present budgets with the abstention of Sumar, PNV, and Junts, without certainty that they can be approved (Sentiment 4, Uncertainty 9).

Sánchez reiterates his commitment to presenting a budget this year and works with parliamentary groups (Sentiment 7, Uncertainty 7).

Thousands of people demonstrate in the three Basque capitals for decent pensions (Sentiment 6, Uncertainty 7).

The pension reform passes the first AIREF exam, which warns of sustainability (Sentiment 7, Uncertainty 8).

Paula Conthe: The improvement in the risk premium is here to stay (Sentiment 7, Uncertainty 6).

Percentage of false positives in each topic

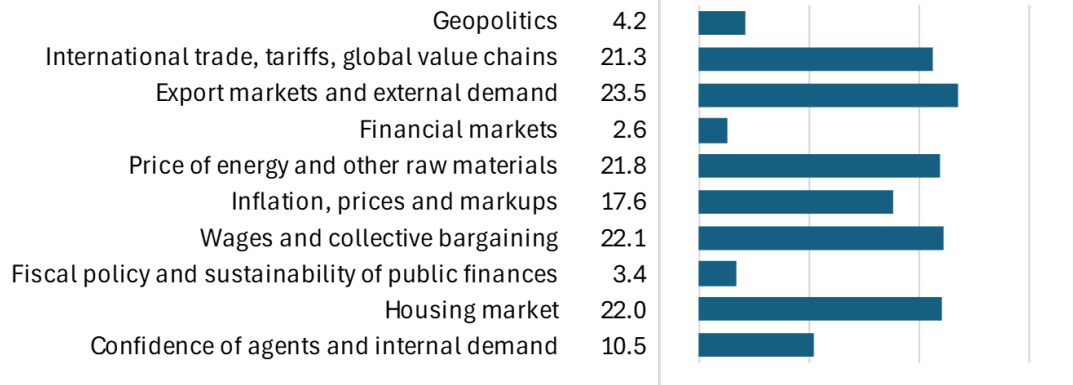


Figure 4: Percentage of false positives in each topic, according to GPT-4o-mini.

The Government begins negotiations on the General Budgets (Sentiment 6, Uncertainty 5).

This is the news article to evaluate:

The prompts are executed via API calls to a private instance of GPT-4o-mini hosted on Azure AI Forge, an environment that complies with the access and confidentiality conditions associated with our news database. This setup ensures that all processing remains within a secure and institutionally approved infrastructure. For each topic, the complete inference process, covering approximately 130,000 articles, requires about 20 hours of computation and entails a total cost of roughly €40, reflecting a favorable balance between accuracy, scalability, and resource efficiency.

As explained in the previous subsection, in addition to requesting the quantification of uncertainty, the prompt explicitly asks the model to confirm whether the article indeed pertains to the relevant topic. This step serves as an additional quality control layer, filtering out potential false positives that may have been introduced during the semantic retrieval phase. The responses provided by the LLM confirm that, in general terms, the news articles retrieved by the embedding model are appropriately aligned with their intended topics. The proportion of false positives identified at this stage, as reported for different topics in Figure 4, remains low: approximately 4 percent for the most clearly delineated topics and up to 20 percent for the most conceptually diffuse ones. This attests to the robustness of the semantic retrieval procedure and to the effectiveness of the subsequent validation step performed by the LLM.

The outputs produced by GPT-4o-mini exhibit a high degree of internal consistency. When the model is prompted a second time with the same set of news articles for a given topic, the correlation between the two rounds of responses ranges between 0.8 and 0.9 at the level

of individual articles, and between 0.98 and 0.99 when looking at monthly averages. These results indicate that, despite the stochastic nature of LLM inference, the evaluation procedure yields highly stable estimates once aggregated over time or across large samples. The examples included in the prompt are an important ingredient for achieving this level of consistency. Figures A3 and A4 in the Appendix illustrate the results of this robustness exercise.

3 Indicators

3.1 Constructed RAUIs

Once the uncertainty scores for each news article are obtained from the LLM, we proceed to construct the topic-specific and aggregate uncertainty indicators. For each of the ten topics, we compute the monthly average of the uncertainty values corresponding to the articles that were both identified and confirmed as belonging to that topic. These averages provide a measure of the perceived uncertainty associated with each topic over time.

To ensure comparability across topics, each series is standardized by subtracting its mean and dividing by its standard deviation. This normalization step allows the resulting indicators to be expressed in standardized units, facilitating interpretation and aggregation. The outcome of this step is a set of ten normalized uncertainty series, each representing the evolution of measured uncertainty in a specific thematic dimension of the news corpus. However, these topic-specific indicators do not yet constitute our final Retrieval-Augmented Uncertainty Indicators (RAUI). Instead, each topic-level RAUI is the product of two complementary components:

- a signal, represented by the level of uncertainty expressed in the news articles for that topic; and
- an amplifier, represented by the frequency or relative weight of that topic in the media.

Both dimensions are essential to capture the dynamics of uncertainty as perceived through the news. The case of geopolitical tensions provides an illustrative example. The average uncertainty expressed in articles classified under this topic in 2016 (many of which discussed conflicts such as the Syrian civil war) was not substantially different from that in 2022 (when articles also discussed the invasion of Ukraine). However, the relative prevalence of this topic increased approximately threefold between these two dates, as the one in Ukraine is a conflict that, from a Spanish perspective, was broadly perceived as "closer". Consequently, the topic's contribution to aggregate uncertainty (which we will define shortly) rose sharply, reflecting

the heightened prominence of geopolitical risk in public discourse even when the tone in each individual article did not change as markedly.

Figures ?? and ?? present the topic-specific results of the analysis. The columns in these figures display three complementary series for each of the ten topics: (i) the average uncertainty level estimated from the articles associated with that topic (the signal); (ii) the relative prevalence of the topic over time, measured as the share of articles classified under it relative to the total number of articles identified as belonging to any of the ten topics in that month (the amplifier); and (iii) the resulting topic-level RAUI, defined as the product of the two. For ease of interpretation of the units in this third column, these topic-level RAUI series have been normalized once again by dividing by their 2000-2024 standard deviation. Since the resulting monthly series is somewhat noisy, a 3-month moving average of the topic RAUI is presented in these figures.

To capture the overall level of uncertainty, we construct an aggregate RAUI as a weighted average of the ten normalized topic-specific uncertainty signals (not the topic RAUI, but the underlying signal series). The weight assigned to each topic in a given month corresponds to its relative prevalence, defined as the share of articles classified under that topic within the total number of identified articles for that month (i.e., the amplifier). This weighting scheme ensures that the aggregate indicator reflects both the intensity of uncertainty within topics and the degree of media attention they receive at each point in time. If we do not apply the final normalization step to each topic's RAUI (the last step used when preparing the data for the third column of Figures ?? and ??, which facilitates interpretation of units), then the aggregate RAUI is simply the sum of the ten topic-level RAUIs. For easier interpretation of the units, this overall RAUI is finally also normalized, dividing by its 2000-2024 standard deviation.

$$RAUI_i = \frac{signal_i * amplifier_i}{stdev(signal_i * amplifier_i)}$$

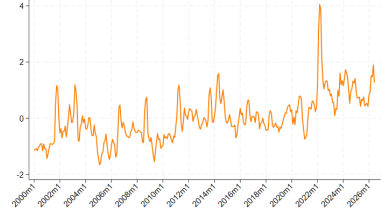
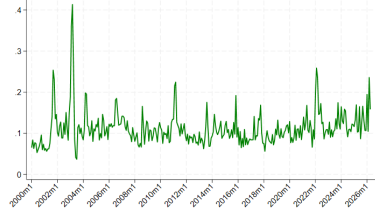
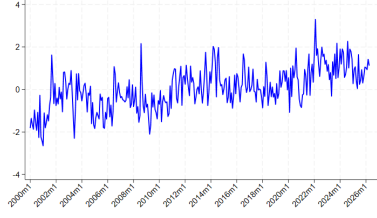
$$RAUI = \frac{\sum_i signal_i * amplifier_i}{stdev(\sum_i signal_i * amplifier_i)}$$

Figure 7 displays the aggregate RAUI, summarizing overall uncertainty across all topics. The indicator behaves as expected for a news-based measure of uncertainty: it rises sharply during the 2001-2003 and 2008-2012 crises, and in episodes such as the outbreak of COVID-19 in 2020, the Russian invasion of Ukraine in 2022, and the introduction of the Trump tariffs in

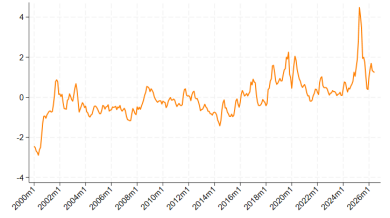
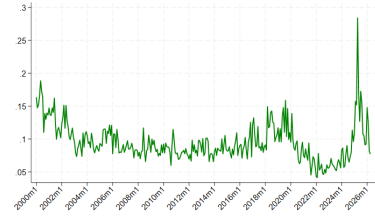
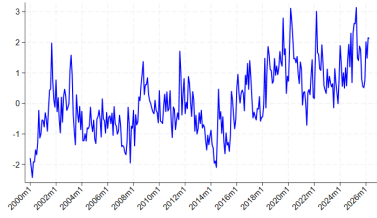
Uncertainty Evaluation (Signal)

Relative Weight (Amplifier)

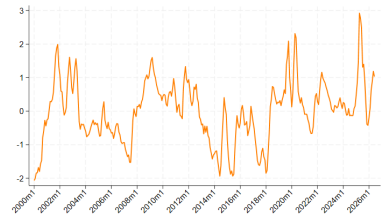
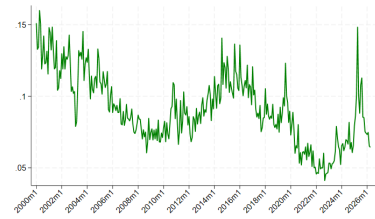
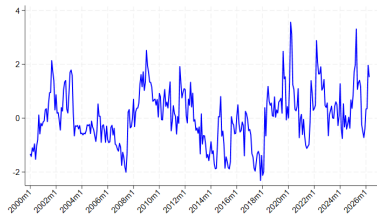
RAUI (3-month moving average)



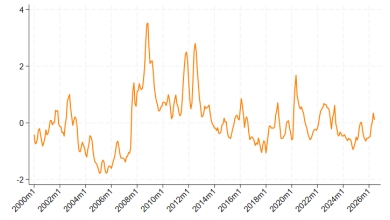
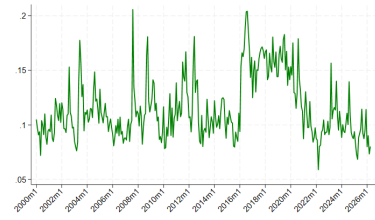
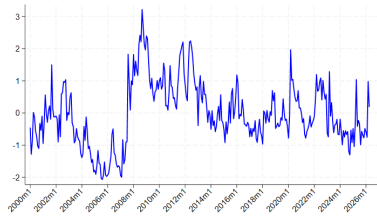
(a) Geopolitical tensions



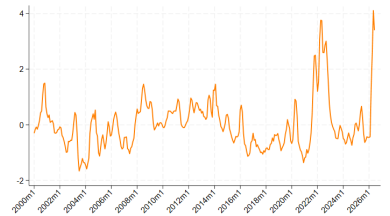
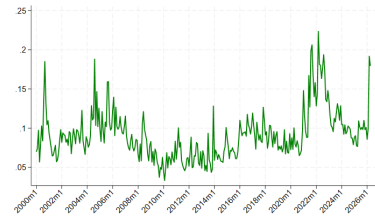
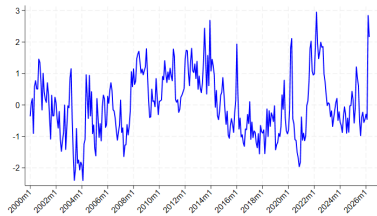
(b) International trade



(c) Export markets



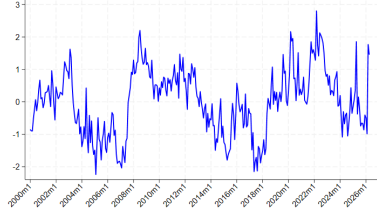
(d) Financial markets



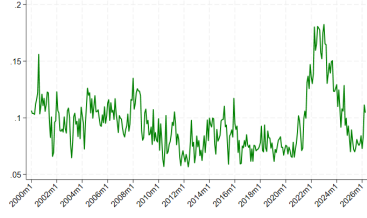
(e) Energy prices

Figure 5: Topic-specific RAUIs and their signal and amplifier components (2000M1–2026M4).

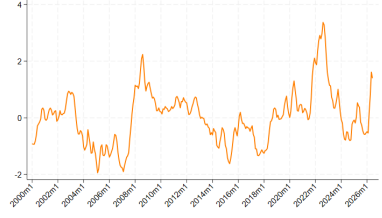
Uncertainty Evaluation (Signal)



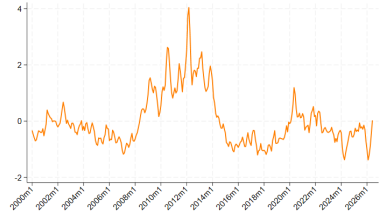
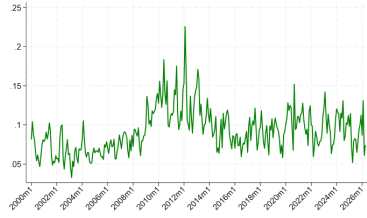
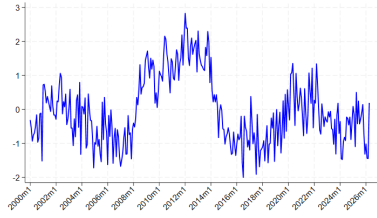
Relative Weight (Amplifier)



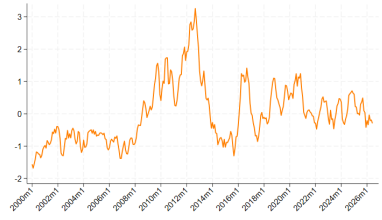
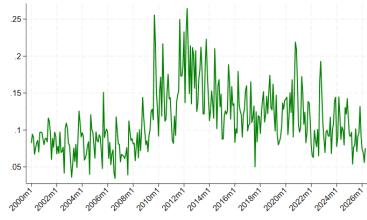
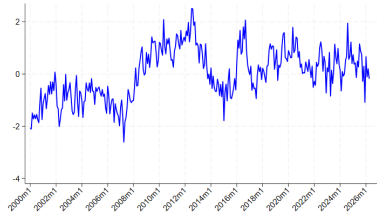
RAUI (3-month moving average)



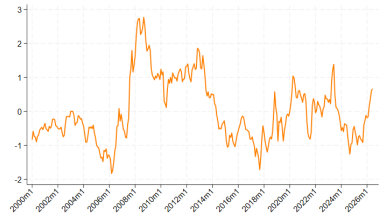
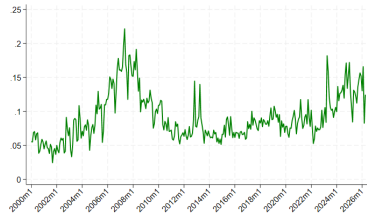
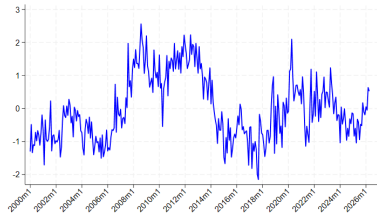
(f) Inflation and prices



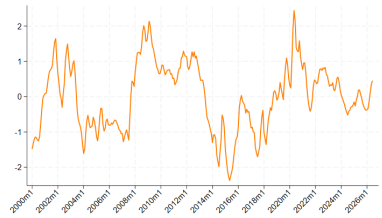
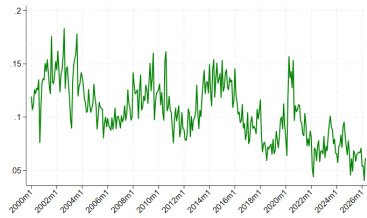
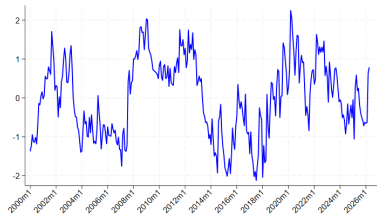
(g) Wages



(h) Fiscal policy



(i) Housing market



(j) Agents' confidence

Figure 6: Topic-specific RAUIs and their signal and amplifier components (2000M1–2026M4).

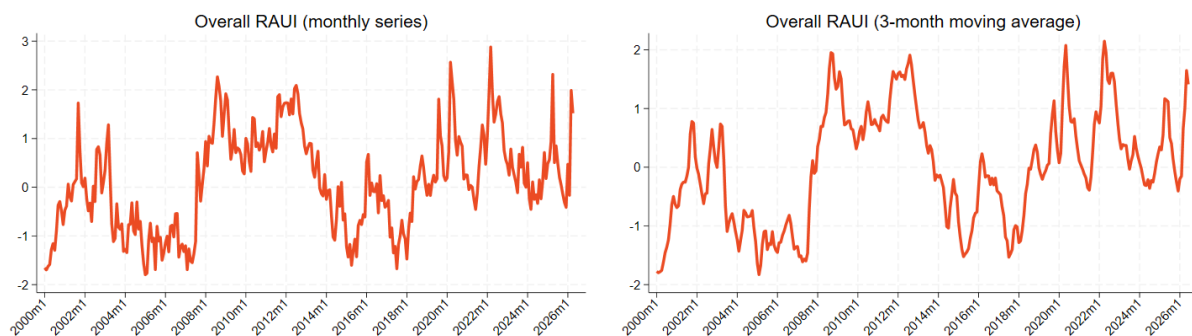


Figure 7: Overall RAUI for Spain: original and smoothed series (2000M1-2026M4)

2025.

At the same time, Figure 7 also illustrates an important limitation of these indicators. In 2007, immediately prior to the onset of the global financial crisis, the aggregate RAUI reached one of its lowest points in the entire sample. This occurs because the indicator captures perceived uncertainty as reflected in contemporary news coverage, rather than the true underlying uncertainty that would, in retrospect, characterize the pre-crisis environment.

Figure 8 provides additional detail by presenting a contribution chart that illustrates how each topic contributes to total uncertainty over time: a key advantage of the RAUI methodology relative to traditional approaches. The chart reveals that during the 2002–2003 episode, the main drivers of heightened uncertainty were factors external to the Spanish economy, notably international trade, geopolitics, and global demand. In contrast, during the 2008–2013 crisis, uncertainty was driven predominantly by domestic factors: housing, financial markets, and prices in the early phase of the crisis, followed by fiscal policy, wages, and financial markets in the latter phase.

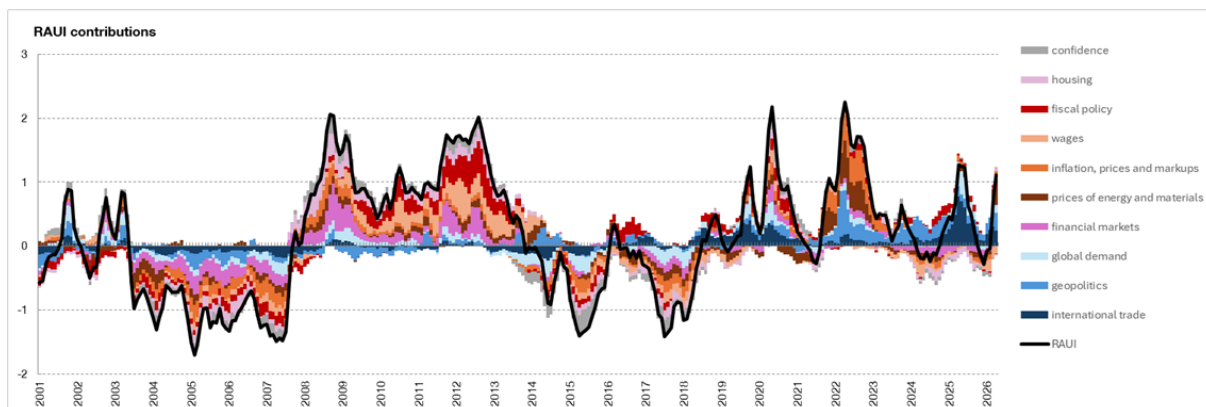


Figure 8: Contributions of individual topics to aggregate RAUI in the full sample. Series are reported as three-month moving averages of monthly contributions.

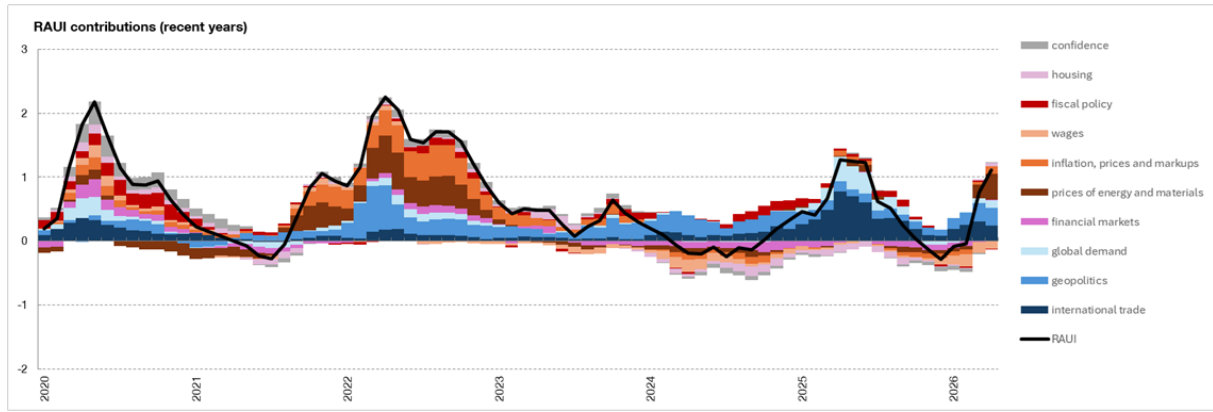


Figure 9: Contributions of individual topics to aggregate RAUI in the post-pandemic period. Series are reported as three-month moving averages of monthly contributions.

Figure 9 presents the same decomposition as Figure 8 but focuses on the six most recent years. With the arrival of COVID-19, all ten topics exhibit a sharp increase in uncertainty. The largest contributions come from international trade (reflecting early-pandemic disruptions to global value chains), financial markets, global demand, and fiscal policy. In late 2021, when tensions between Russia and Ukraine started to appear, there was an initial spike in uncertainty driven primarily by the two topics related to prices and inflation; this was followed by a further rise in early 2022 when the Russian invasion started, and the topic of geopolitics became a major contributor. In 2025, another surge in uncertainty appears, this time led by international trade and global demand, reflecting the introduction of the Trump tariffs.

3.2 Comparison with Other Uncertainty Indicators

Two key features of RAUI are, first, that it is topic-based and allows us to decompose the drivers of aggregate uncertainty at each point in time, and second, that it combines a sophisticated assessment of the level of uncertainty within each article (the signal, calculated with the LLM) with an equally sophisticated frequency measure (the amplifier, provided by the semantic search with an embedding model). This combination yields indicators that capture both the relevance of a specific topic and the magnitude of uncertainty expressed in the news over time. Figure 10 compares the evolution of RAUI with two widely used measures of uncertainty for the Spanish economy. The first is the Economic Policy Uncertainty (EPU) index constructed by Ghirelli et al. (2019), which is based on counting policy uncertainty-related terms in Spanish news articles. The second is the Financial Volatility Index (VIBEX), which measures expected short-horizon volatility of the IBEX 35 stock index and is derived from option prices.

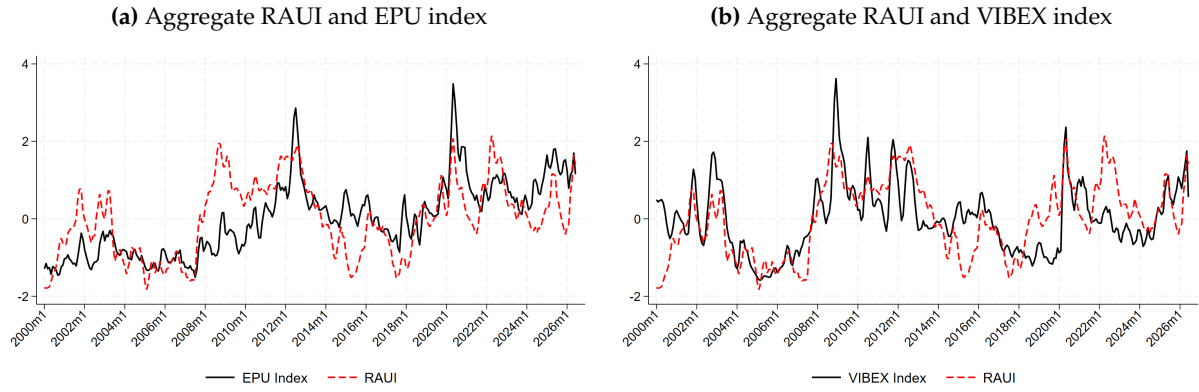


Figure 10: Comparison of aggregate RAUI with the Economic Policy Uncertainty (EPU) index constructed by [Ghirelli et al. \(2019\)](#) and the Financial Volatility Index (VIBEX). Series are reported as three-month moving averages of the monthly indicators.

Although the three indicators are positively correlated, each captures different facets of aggregate uncertainty. For example, the EPU index remains relatively muted in 2001–2002 and in the early stages of the Global Financial Crisis in 2008, despite significant uncertainty stemming from geopolitical tensions and financial market turmoil in those periods. On the other hand, the VIBEX primarily reflects uncertainty priced in financial markets and therefore falls short of capturing episodes driven by broader macroeconomic or geopolitical shocks, such as the rise in uncertainty in Spain in 2018 or following Russia’s invasion of Ukraine in 2022. Table 1 reports the pairwise correlations between aggregate RAUI and these two alternatives, showing correlations of 0.56 with VIBEX and 0.58 with EPU. Notably, VIBEX exhibits a strong correlation with the RAUI for financial markets (0.69), while EPU is correlated with the RAUI for fiscal policy and debt sustainability (0.55). This pattern underscores that standard uncertainty proxies tend to emphasize specific dimensions of uncertainty (related to economic policy and to financial markets, in these examples), whereas RAUI provides a more comprehensive measure that can also be flexibly adapted to incorporate new topics if future developments require it.

	RAUI	RAUI-GP	RAUI-FM	RAUI-FP	GPR Index	VIBEX Index	EPU Index
RAUI	1.00						
RAUI-GP	0.40***	1.00					
RAUI-FM	0.77***	0.11	1.00				
RAUI-FP	0.65***	0.16**	0.48***	1.00			
GPR Index	-0.03	0.16**	-0.13*	-0.14*	1.00		
VIBEX Index	0.56***	0.06	0.69***	0.29***	0.03	1.00	
EPU Index	0.58***	0.41***	0.39***	0.55***	-0.03	0.41***	1.00
N	316						

Table 1: Correlations between aggregate RAUI and its geopolitical (RAUI-GP), financial markets (RAUI-FM), and fiscal policy (RAUI-FP) components, and three external uncertainty measures for Spain: the Geopolitical Risk (GPR) index developed by [Caldara and Iacoviello \(2022\)](#), the Financial Volatility Index (VIBEX), and the Economic Policy Uncertainty (EPU) index constructed by [Ghirelli et al. \(2019\)](#).

Additionally, by incorporating both the frequency and the content of news coverage, RAUI captures a notion of uncertainty that is closer to how economic agents perceive and process uncertainty, rather than relying on either dimension alone. Figure 11 compares the dictionary-based Geopolitical Risk index (GPR) for Spain developed by [Caldara and Iacoviello \(2022\)](#) with the geopolitical component of RAUI and its subcomponents. As shown in the figure and summarized in Table A2 in the Appendix, the amplifier component of RAUI-GP is strongly correlated with the GPR index ($\rho = 0.50$), reflecting the fact that both measures are primarily driven by the relative frequency of news related to geopolitical tensions. By contrast, the correlation with the overall RAUI-GP measure is substantially weaker ($\rho = 0.16$), due to limited comovement with the signal component. In particular, while the GPR index for Spain peaks in 2004, RAUI-GP reaches its highest levels in 2022 and indicates comparatively lower geopolitical uncertainty in the aftermath of the 2004 Madrid terrorist attacks.

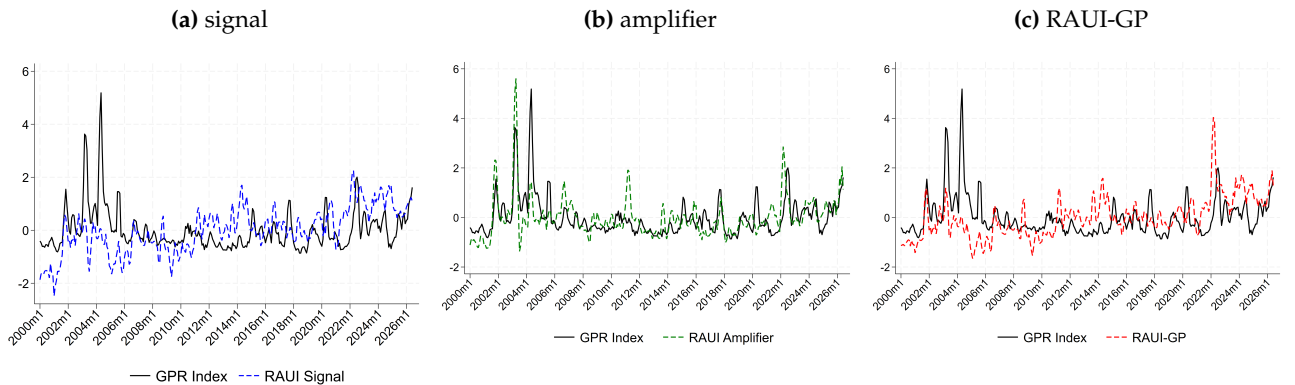


Figure 11: Comparison of the Geopolitical Risk (GPR) index and the RAUI for geopolitical tensions. Series are reported as three-month moving averages of the monthly indicators.

As mentioned above, the VIBEX index, which captures how Spanish financial markets price uncertainty, exhibits a strong correlation with our financial-markets uncertainty measure (RAUI-FM), with $\rho = 0.69$. This comovement is driven almost entirely by the signal component of RAUI-FM ($\rho = 0.67$), while the correlation with the amplifier component is essentially zero. Figure 12 illustrates this pronounced co-movement between RAUI-FM and VIBEX, highlighting that it originates primarily from the uncertainty-scoring (signal) dimension. In contrast, relying solely on the relative frequency of news coverage related to financial markets (the amplifier component) yields a markedly different time profile: it understates financial uncertainty during the global financial crisis and overstates uncertainty in the pre-COVID years.

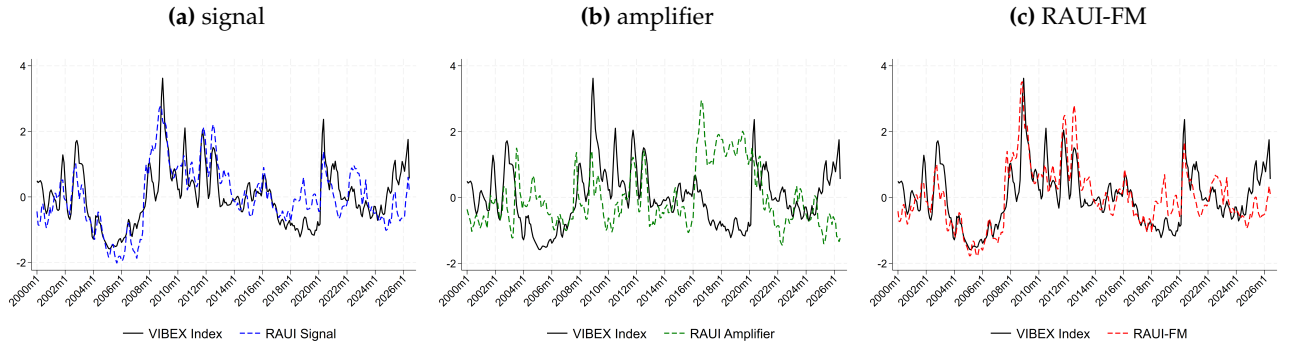


Figure 12: Comparison of the VIBEX index and the RAUI for financial markets. Series are reported as three-month moving averages of the monthly indicators.

4 Application: Macroeconomic Effects of Uncertainty

In this section, we use our constructed uncertainty measures to analyze the macroeconomic implications of uncertainty shocks using a vector autoregression (VAR) model. To this end, we estimate a VAR using our monthly dataset spanning January 2000 to December 2025. The baseline specification includes six variables: (i) the uncertainty index, (ii) the log of the Industrial Production Index (IP), (iii) the unemployment rate, (iv) the log of the IBEX stock market index, (v) the annual growth rate of the Consumer Price Index (CPI), and (vi) the overnight interbank interest rate⁶. The baseline model is estimated with six lags. We identify uncertainty shocks using a Cholesky decomposition of the covariance matrix of the reduced-form VAR residuals. The uncertainty measure is ordered first, implying that any contemporaneous correlation

⁶A detailed description of the data sources and summary statistics is provided in Appendix Section A. Additionally, we include three time-dummy variables for 2020M3, 2020M4, and 2020M5 in the VAR specification to ensure that the extreme observations during the onset of the COVID-19 pandemic do not distort the results.

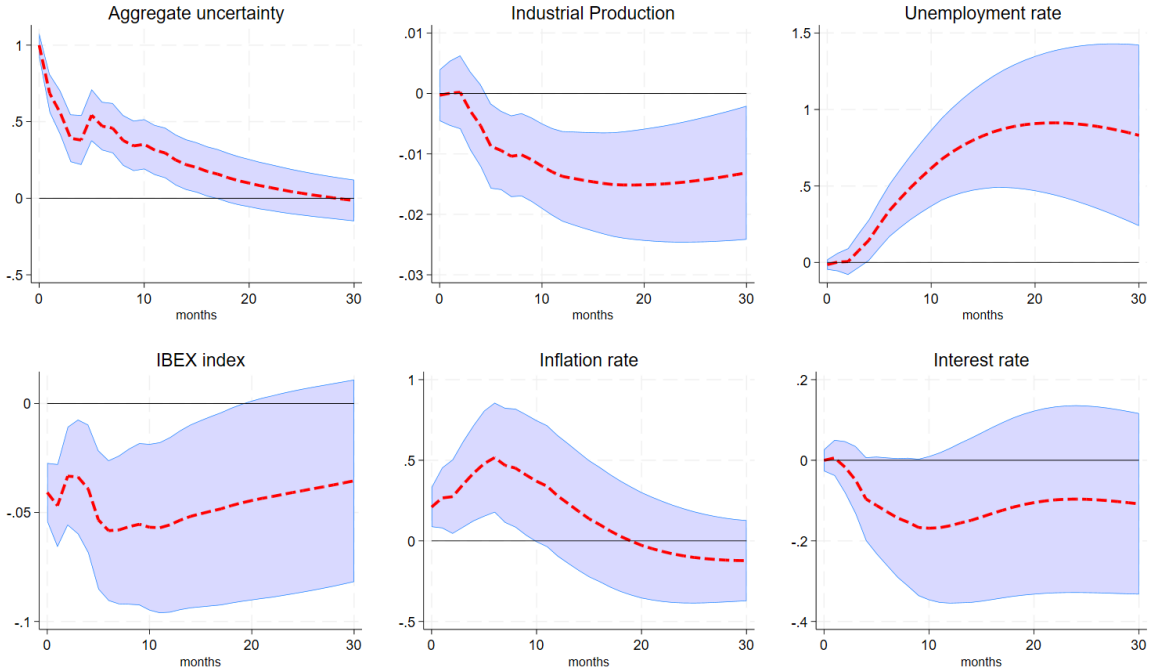


Figure 13: Impulse responses of all variables to a one standard deviation shock to the aggregate uncertainty index. The sample period spans from 2000M1 to 2025M12. Shaded areas represent 95% confidence intervals.

between uncertainty and macroeconomic variables is attributed to the effect of uncertainty on those variables. This recursive identification strategy is commonly employed in the literature to isolate uncertainty shocks, and its plausibility has been discussed in several influential studies (see, for example, [Baker et al. \(2016\)](#) and [Caldara and Iacoviello \(2022\)](#), among others)⁷.

Impulse response functions (IRFs) to a one-standard-deviation innovation in the uncertainty index are presented in Figure 13. Following a one-standard-deviation increase relative to the mean, the uncertainty index remains persistently elevated for up to 20 months. In response to the shock, industrial production begins to decline after two months, reaching a peak contraction of approximately 1.5 percent around 18 months post-shock. The unemployment rate rises gradually, peaking at a 0.8 percentage point increase around the same horizon. The IBEX stock index experiences a sharp and immediate decline, indicating a swift adverse reaction from financial markets. Inflation responds with a temporary spike that dissipates within a year, while the interest rate shows a sustained decline, consistent with monetary policy accommodation in response to weakening economic activity.

The results in Figure 13 show that, consistent with much of the existing literature, an un-

⁷As a robustness check, we order the uncertainty measure last in the VAR, allowing it to respond contemporaneously to the other variables. Figure A5 in the Appendix shows that the results are similar to the baseline specification.

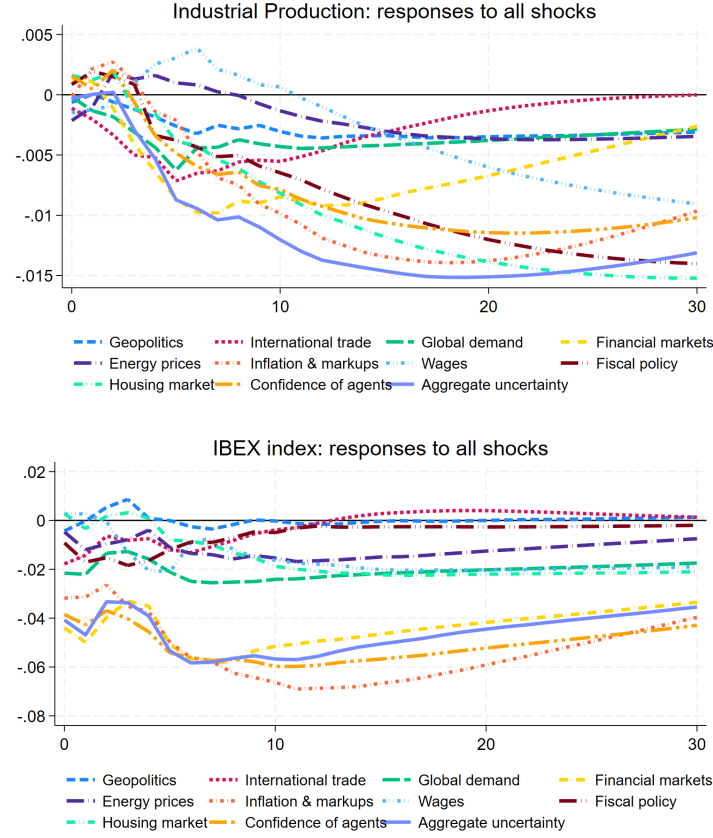


Figure 14: Impulse responses of industrial production and stock market index to a one standard deviation shock to different uncertainty measures. The sample period spans from 2000M1 to 2025M12.

certainty shock exerts a significant adverse effect on both real economic activity and financial markets. A key advantage of our framework lies in its flexibility to construct topic-specific uncertainty indices, allowing for a more granular examination of the sources driving the aggregate uncertainty–macro relationship. To this end, we replace the aggregate index with each of the ten individual uncertainty measures, re-estimating the VAR while keeping the rest of the specification unchanged. Figures A6 to A15 in the Appendix display the results of this exercise for each topic-based uncertainty measure and reveal significant heterogeneity in their dynamic responses.

Figure 14 summarizes the results of this exercise by displaying the impulse responses of industrial production and the stock market index across different types of uncertainty shocks. The corresponding responses for the remaining macroeconomic variables are presented in Figure A16 in the Appendix. While all indices broadly exhibit contractionary effects on industrial production and the stock market, there is considerable variation in the timing and magnitude of the responses. Uncertainty measures related to inflation and markups, fiscal policy, housing markets, internal demand, and financial markets generate more pronounced macroeco-

nomic effects. In contrast, measures linked to international sources, such as geopolitical risk and global trade, produce more muted responses, underscoring the comparatively limited exposure of the Spanish economy to external uncertainty shocks. These findings highlight the heterogeneous transmission mechanisms associated with different forms of uncertainty and suggest that aggregate uncertainty captures a composite of distinct dynamics originating from diverse sources ⁸. Next, we employ principal component analysis (PCA) to extract the main common components underlying our uncertainty measures. This allows us to isolate the dominant drivers of fluctuations in uncertainty and assess how they interact and transmit through the macroeconomy.

The results of the principal component analysis, summarized in Tables A3 and A4 in the Appendix, indicate that the first two components account for approximately two-thirds of the total variation in the uncertainty measures: 46% and 16% of the variance are explained by the first and second components, respectively. The first component (PC1) predominantly reflects domestic macro-financial uncertainty, with high loadings on indices related to agents' confidence and internal demand, housing, inflation and markups, financial markets, wages, and fiscal policy. In contrast, the second component (PC2) primarily captures international sources of uncertainty, such as geopolitical risk and international trade uncertainty. Global demand uncertainty contributes meaningfully to both components, suggesting its role as a shared driver of both domestic and international uncertainty dynamics.

Figure 15 illustrates the time series of PC1 and PC2, revealing the evolution of the main underlying forces of uncertainty in the Spanish economy. The figure shows that while the two components often co-move, such as during the early 2000s and in 2022, they also diverge notably in specific periods. For example, during the European sovereign debt crisis, domestic uncertainty remained elevated while international uncertainty was relatively subdued. Conversely, in the final months of 2024, international uncertainty rose markedly ahead of the U.S. presidential election, whereas domestic uncertainty began to rise only with a lag of several months. These dynamics highlight the importance of disentangling distinct sources of uncertainty when analyzing macroeconomic responses.

We re-estimate the VAR specification by replacing the aggregate uncertainty index with PC2 and PC1, entering them jointly as the first two variables in the Cholesky ordering. This ordering is motivated by the assumption that uncertainty stemming from international factors is

⁸Appendix Figures A17, A18, and A19 further compare our results with the Economic Policy Uncertainty (EPU) index, the Geopolitical Risk (GPR) index for Spain, and the financial volatility index (VIBEX).

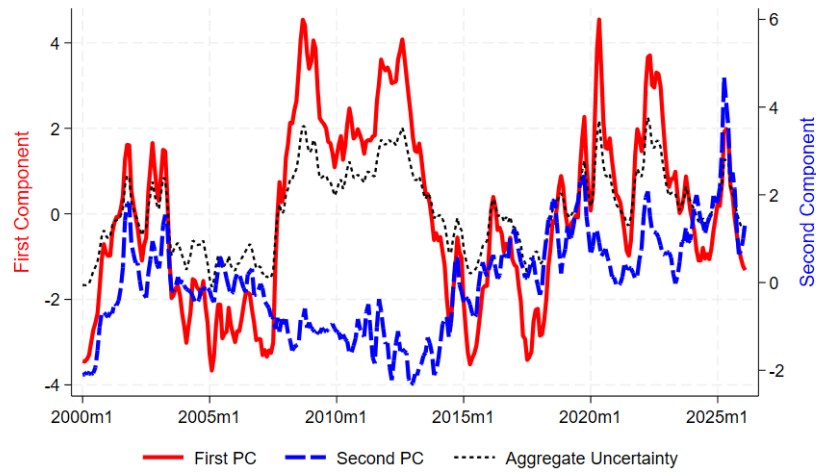


Figure 15: First and second principal components of the uncertainty measures. The solid red line represents the first (internal) component, the dashed blue line corresponds to the second (international) component, and the dotted black line depicts the aggregate uncertainty index. The sample period spans from 2000M1 to 2025M12.

contemporaneously exogenous to domestic macroeconomic variables. The remaining macroeconomic variables are ordered as before. The impulse response results from this specification are shown in Figures 16 and 17. A positive shock to international uncertainty (PC2) increases the internal uncertainty drivers and generates a temporary, short-lived effect on output, which reaches a peak contraction of 0.9 percent and dissipates after ten months. In response to this shock, the stock market also shows a short-lived drop, whereas the effects on unemployment and inflation are more muted and often statistically insignificant.

In contrast, a shock to internal uncertainty (PC1), shown in Figure 17, produces larger and more persistent macroeconomic effects. Industrial production shows a sustained decline, with a peak contraction exceeding 1 percent after two years. The stock market index responds more sharply and persistently relative to a PC2 shock, and the unemployment rate rises steadily, reaching an increase of up to 0.8 percentage points. Inflation also rises for up to one year. The interest rate response to domestic uncertainty shocks is relatively muted, in contrast to the negative and statistically significant response observed after international uncertainty shocks, consistent with the latter depressing output at the European level. In summary, these results suggest that internal sources of uncertainty account for the primary macroeconomic impact embedded in the aggregate uncertainty measure, whereas international factors exert more limited and transitory effects.

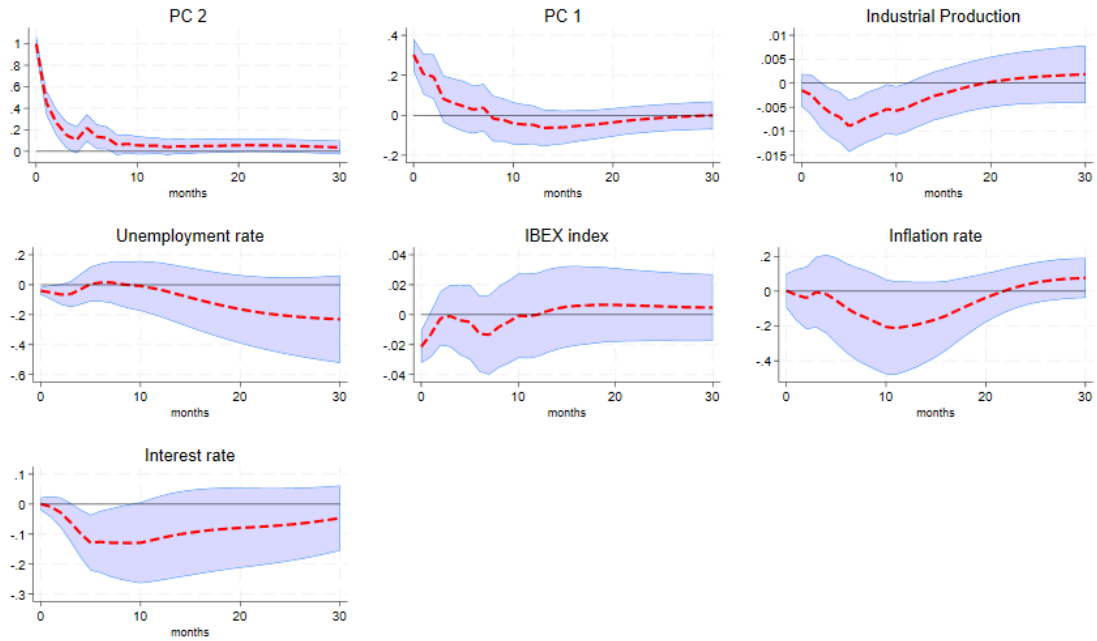


Figure 16: Impulse responses of all variables to a one standard deviation shock to the international component of the uncertainty measures. The sample period spans from 2000M1 to 2025M12. Shaded areas represent 95% confidence intervals.

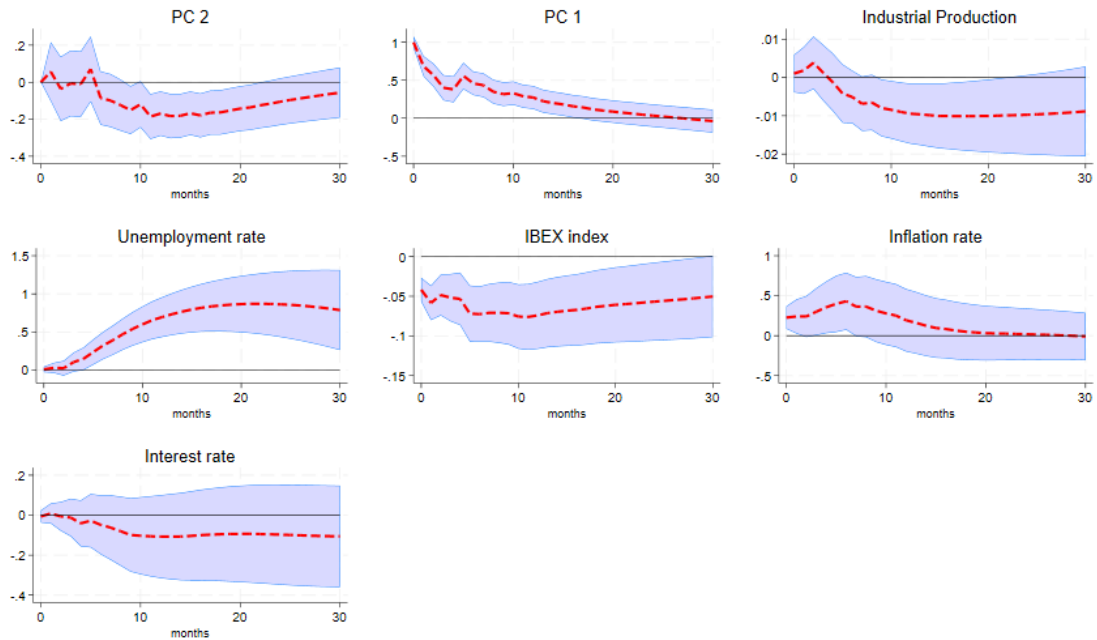


Figure 17: Impulse responses of all variables to a one standard deviation shock to the internal component of the uncertainty measures. The sample period spans from 2000M1 to 2025M12. Shaded areas represent 95% confidence intervals.

5 Conclusion

In this paper, we propose a new methodology for measuring topic-specific economic uncertainty using textual data from newspaper articles, improving on traditional dictionary-based approaches. Word-count methods, while transparent and easy to apply, struggle to capture contextual meaning and can misclassify sentiment or uncertainty when language expressions are not straightforward. To address these limitations without relying on large-scale querying of language models, we develop the Retrieval-Augmented Uncertainty Indicator (RAUI), which combines semantic search based on embeddings with targeted evaluation by an LLM.

This approach allows us to efficiently identify relevant news articles and then use an LLM to quantify their level of uncertainty, enabling the construction of indicators for multiple thematic areas and an aggregate measure that decomposes overall uncertainty into topic-level contributions. Applied to Spanish news data, the resulting indicators reveal how uncertainty evolves over time and which topics drive changes in uncertainty at each point.

We provide a practical application of these indicators in a VAR analysis and show that different sources of uncertainty have distinct macroeconomic effects, with internal uncertainty generating larger impacts on domestic output. These results highlight the value of RAUI in both monitoring the economic environment and informing macroeconomic analysis.

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Appendix

A Data Description

A.1 Newspaper Data

The newspaper data in our analysis come from Dow Jones Factiva DNA. The query selects and downloads more than 4 million articles, using the following criteria:

- From 2000 onwards
- The title plus body add up to 100 characters or more
- Published by one of the following 8 Spanish newspapers: ABC, Cinco Dias, El Economista, El Mundo, El Pais, Expansion, La Razon, La Vanguardia.
- Appearing in the following newspaper sections: corporations and industry news, economic news, markets, editorials, government, politics, and international relations. We also include articles outside these sections if Dow Jones tags them as pertaining to politics, consumers, risks, or disasters.

Figure A1 shows the number of articles included in the download with this query, by year and by source.

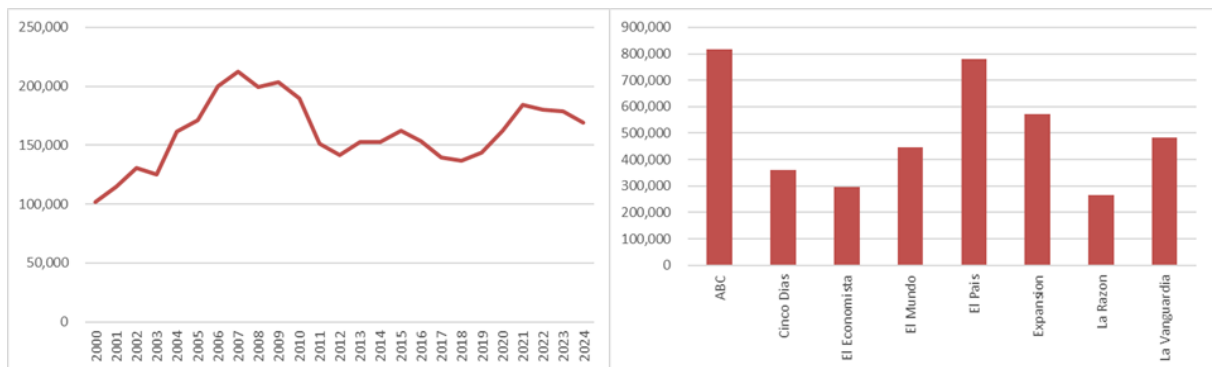


Figure A1: Number of news articles by year and by source.

A.2 Monthly Series

The monthly aggregate data are compiled from multiple sources. The Industrial Production Index and the inflation rate are obtained from the Spanish National Statistics Institute (INE). The unemployment rate is sourced from the Eurostat database. Data on the IBEX stock market index are retrieved from the Banco de España database, while the overnight interest rate is

taken from the Federal Reserve Economic Data (FRED). Summary statistics for these variables are reported in Table A1.

	mean	sd	min	max	N
Aggregate Uncertainty	0.1	1.0	-1.8	2.9	312
Industrial Production	107.2	11.9	68.0	132.7	312
IBEX Index	9814.7	2132.5	5431.7	17307.8	312
Unemployment Rate	15.2	5.3	7.9	26.4	312
Inflation Rate	2.4	2.0	-1.4	10.8	312
Interest Rate	1.4	1.7	-0.6	5.1	312

Table A1: Descriptive Statistics for Monthly Data from 2000M1 to 2025M12

Correlation between RAUI and other indices

	Topic-based RAUI		
	Signal component	Amplifier component	RAUI
GPR	0.12*	0.50***	0.16**
VIBEX	0.67***	-0.03	0.69***

Notes: “Signal” and “Amplifier” are defined index-by-index (i.e., the Signal/Amplifier components differ across GP, and VIBEX). The RAUI column refers to the corresponding topic-specific index (RAUI-GP, RAUI-FM).

Table A2: Correlations between external indices and their corresponding RAUI components

B Robustness Exercises

This section presents additional robustness exercises for the construction of the RAUI indicators. We first examine the performance of alternative embedding models after applying a secondary validation step with GPT-4o-mini to remove false positives. Figure A2 shows the Gini-like index constructed for all models after dropping false-positive news articles.

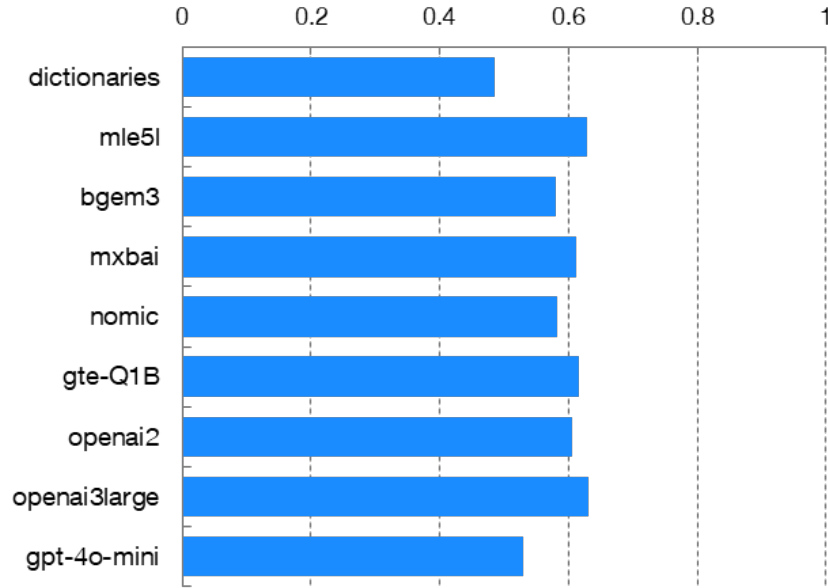


Figure A2: Gini-like index that measures the average performance of each embedding model across the 11 topics, after the secondary check with GPT-4o-mini has been used to eliminate false positives.

We next assess the stability of the uncertainty-evaluation step by comparing the monthly series obtained from two separate executions of the methodology. Figure A3 shows that the resulting series are highly correlated for the first two topics, suggesting that the aggregate indicators are robust to repeated LLM evaluations.

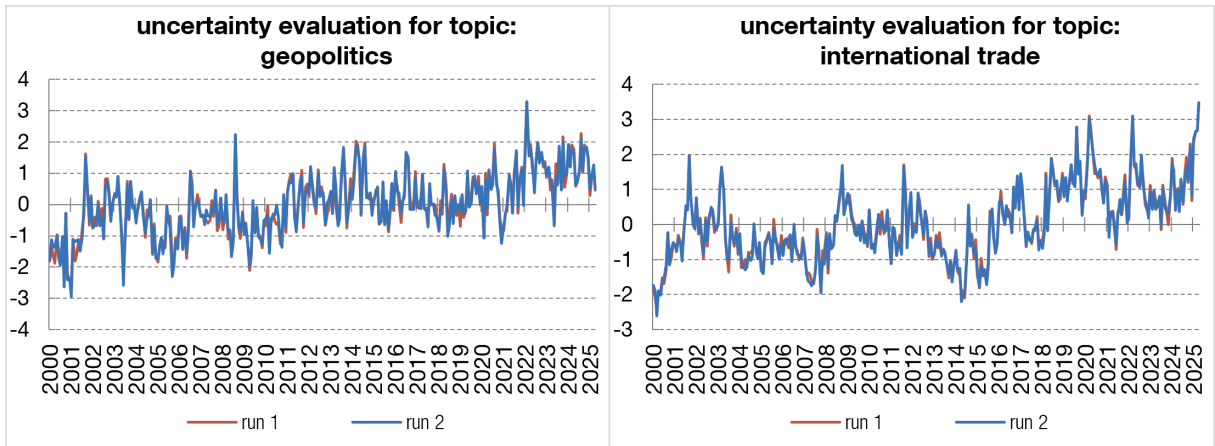


Figure A3: Robustness exercise at the level of monthly series: high correlation between two separate executions of the uncertainty-evaluation methodology for the first two topics.

Figure A4 reports the bivariate distribution of the two scores at the level of individual articles and shows a high degree of consistency across runs.

uncertainty evaluation		run 1											
		0	1	2	3	4	5	6	7	8	9	10	reject
run 2	0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	2	0.0	0.0	0.6	0.5	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.1
	3	0.0	0.0	0.5	2.3	1.3	0.1	0.0	0.0	0.0	0.0	0.0	0.1
	4	0.0	0.0	0.1	1.4	8.4	1.8	0.7	0.1	0.0	0.0	0.0	0.3
	5	0.0	0.0	0.0	0.0	2.0	3.1	2.0	0.7	0.0	0.0	0.0	0.2
	6	0.0	0.0	0.0	0.0	0.7	1.8	6.4	4.5	0.1	0.0	0.0	0.1
	7	0.0	0.0	0.0	0.0	0.2	0.8	5.0	21.9	2.6	0.0	0.0	0.2
	8	0.0	0.0	0.0	0.0	0.0	0.0	0.1	2.4	4.8	0.2	0.0	0.0
	9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.1	0.0	0.0
	10	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	reject	0.0	0.0	0.1	0.1	0.3	0.2	0.2	0.1	0.0	0.0	0.0	20.6

Figure A4: Robustness exercise at the level of individual articles: bivariate distribution of results after two independent GPT-4o-mini evaluations of each of the 130,000 news articles belonging to the second topic. Values are reported in percentages. For example, 1.3% of the 130,000 news articles received an uncertainty evaluation of 4 in the first run and 3 in the second run. The results show a high correlation between the individual answers from the LLM.

C Extra VAR Results

Figure A5 reports the impulse responses to an aggregate uncertainty shock under an alternative recursive identification scheme. In this specification, the uncertainty measure is ordered last in the VAR, allowing it to respond contemporaneously to the other variables in the system.

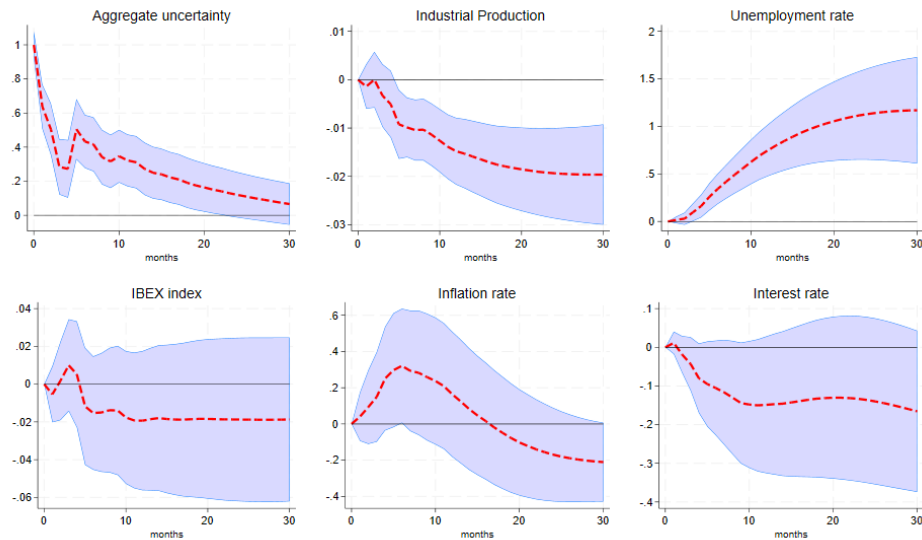


Figure A5: Impulse responses of all variables to a one standard deviation shock to the **aggregate uncertainty**, using the alternative identification scheme. The sample period spans from 2000M1 to 2025M12. Shaded areas represent 95% confidence intervals.

Moreover, we re-estimate the baseline VAR and replace the aggregate uncertainty index

with each of the ten individual uncertainty measures. Figures A6 to A15 display the results of this exercise for each topic-based uncertainty measure. Figure A16 summarizes the results of this exercise by displaying the impulse responses of unemployment rate, inflation, and interest rate across different types of uncertainty shocks. Lastly, Figures A17 and A18 report the results when the aggregate uncertainty index is replaced with the Economic Policy Uncertainty index for Spain constructed by Ghirelli et al. (2019), and the country-specific Geopolitical Risk index developed by Caldara and Iacoviello (2022), respectively.

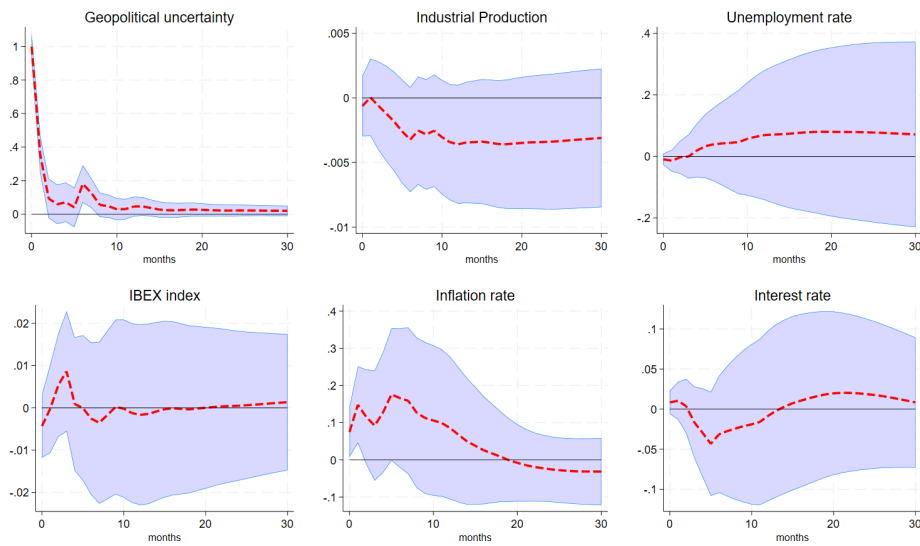


Figure A6: Impulse responses to a one standard deviation shock to **geopolitical uncertainty**. The sample period spans from 2000M1 to 2025M12. Shaded areas represent 95% confidence intervals.

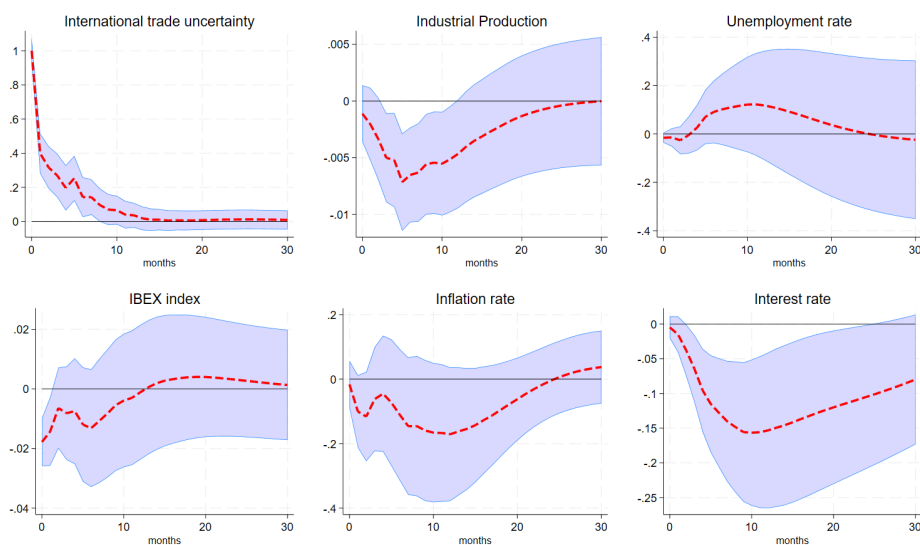


Figure A7: Impulse responses to a one standard deviation shock to **international trade uncertainty**. The sample period spans from 2000M1 to 2025M12. Shaded areas represent 95% confidence intervals.

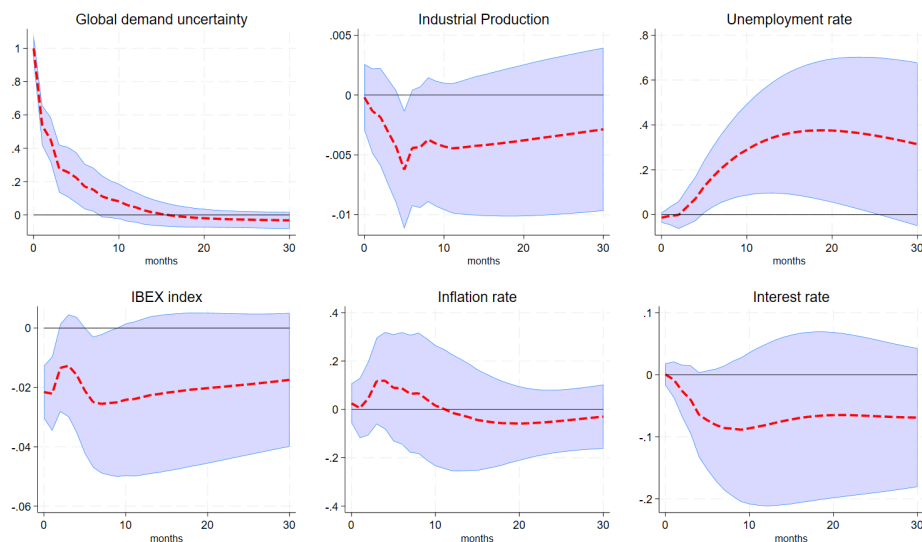


Figure A8: Impulse responses to a one standard deviation shock to **global demand uncertainty**. The sample period spans from 2000M1 to 2025M12. Shaded areas represent 95% confidence intervals.

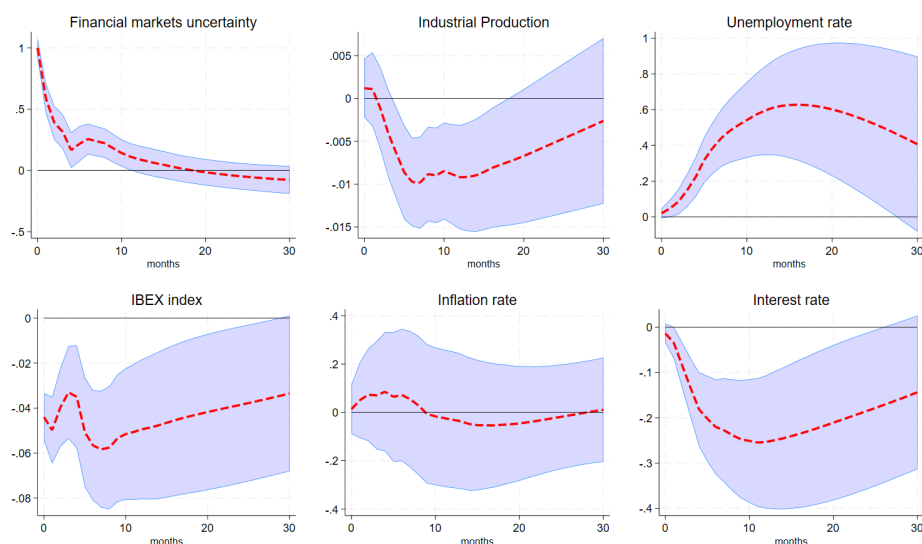


Figure A9: Impulse responses to a one standard deviation shock to **financial markets uncertainty**. The sample period spans from 2000M1 to 2025M12. Shaded areas represent 95% confidence intervals.

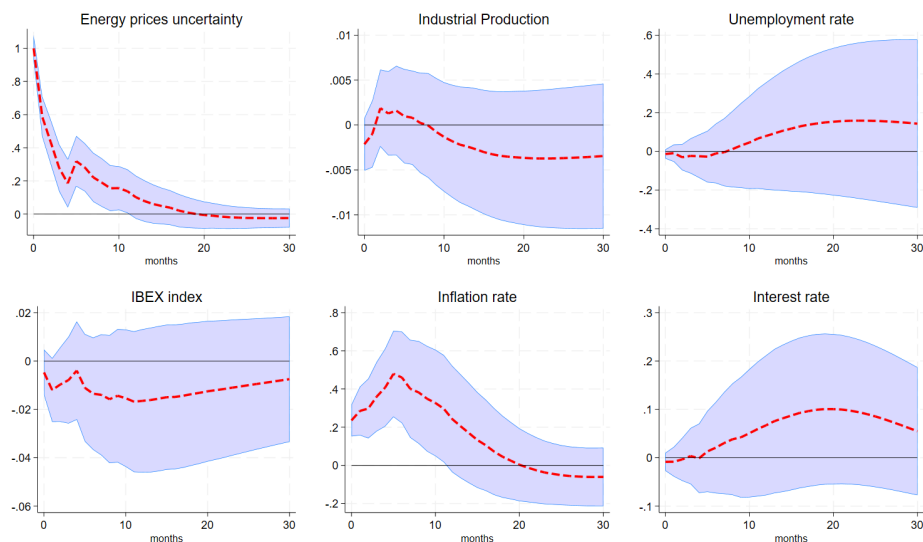


Figure A10: Impulse responses to a one standard deviation shock to **energy prices uncertainty**. The sample period spans from 2000M1 to 2025M12. Shaded areas represent 95% confidence intervals.

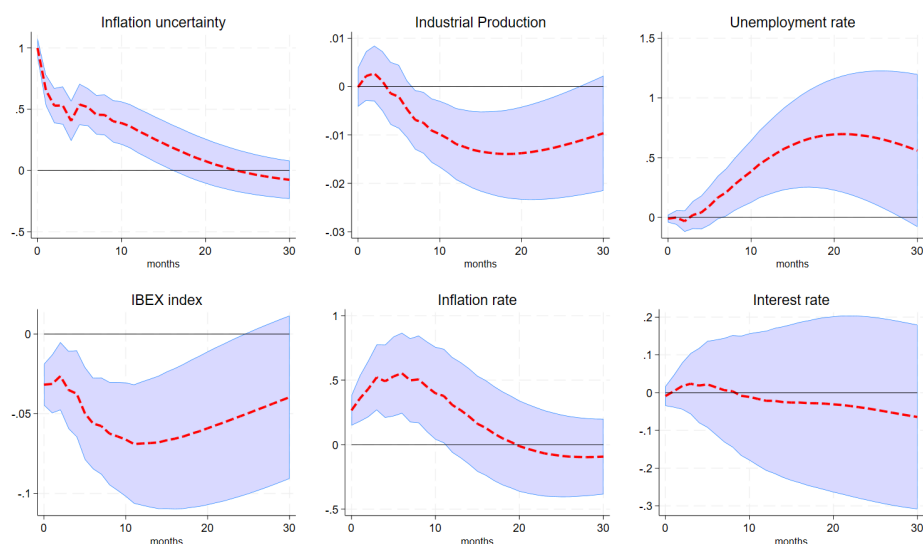


Figure A11: Impulse responses to a one standard deviation shock to **inflation and mark-up uncertainty**. The sample period spans from 2000M1 to 2025M12. Shaded areas represent 95% confidence intervals.

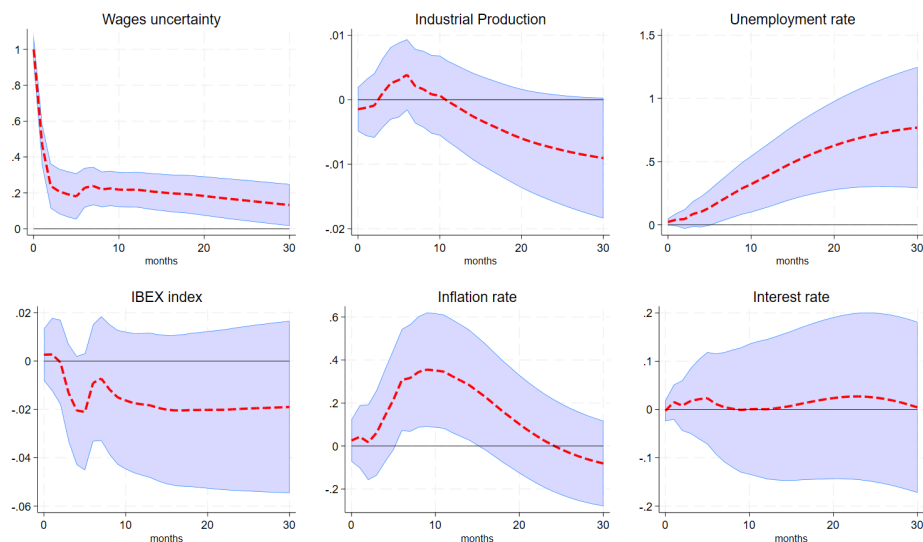


Figure A12: Impulse responses to a one standard deviation shock to **wages uncertainty**. The sample period spans from 2000M1 to 2025M12. Shaded areas represent 95% confidence intervals.

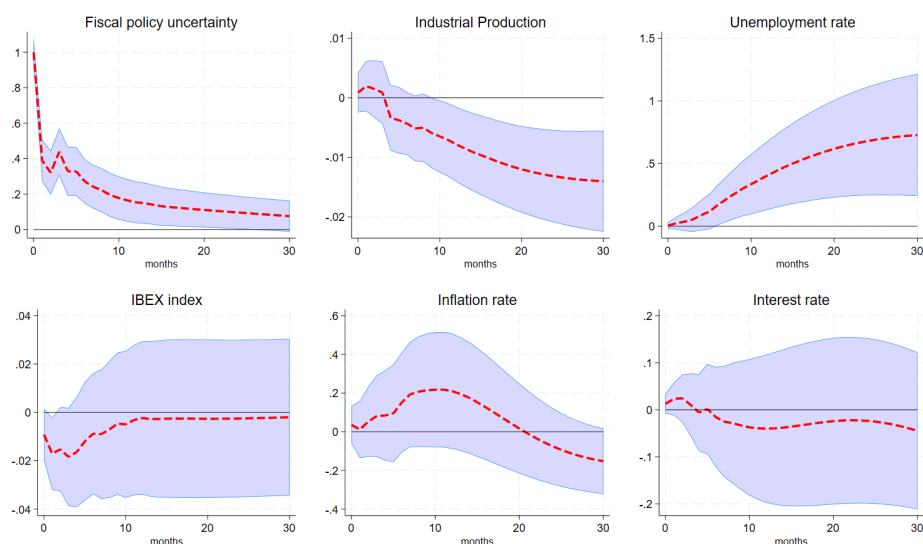


Figure A13: Impulse responses to a one standard deviation shock to **fiscal policy uncertainty**. The sample period spans from 2000M1 to 2025M12. Shaded areas represent 95% confidence intervals.

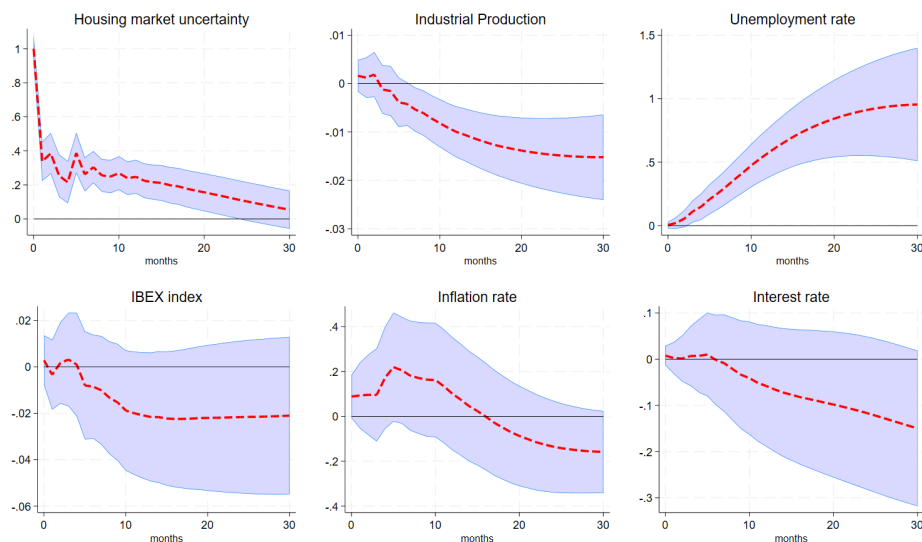


Figure A14: Impulse responses to a one standard deviation shock to **housing market uncertainty**. The sample period spans from 2000M1 to 2025M12. Shaded areas represent 95% confidence intervals.

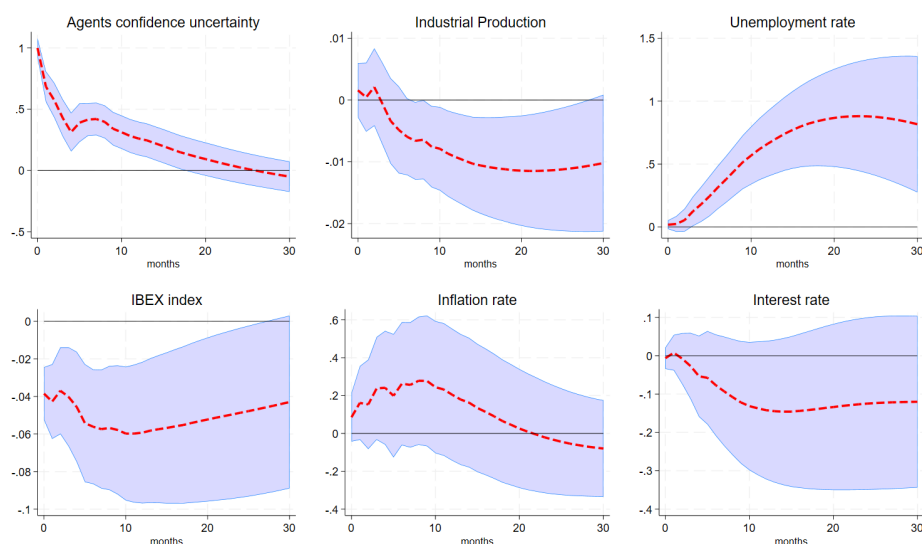


Figure A15: Impulse responses to a one standard deviation shock to **agents' confidence uncertainty**. The sample period spans from 2000M1 to 2025M12. Shaded areas represent 95% confidence intervals.

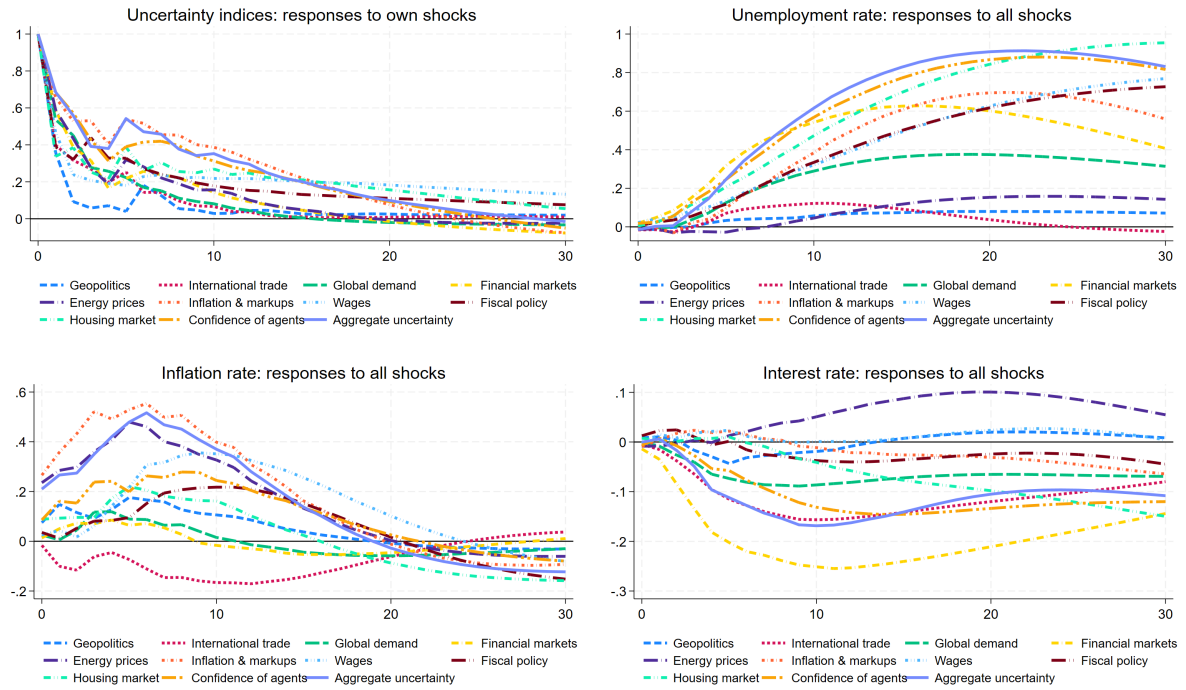


Figure A16: VAR analysis- the response of each variable to a one s.d shock to different uncertainty measures

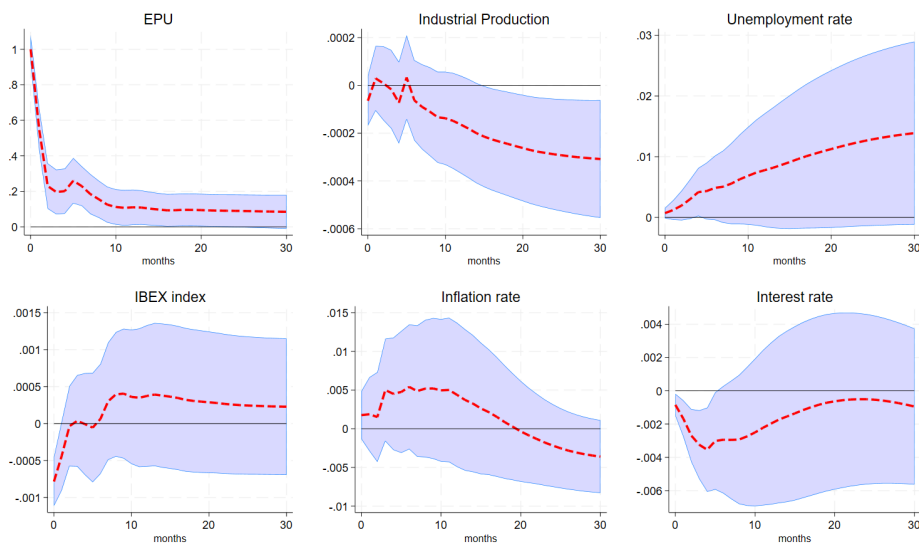


Figure A17: Impulse responses to a one standard deviation shock to the **Economic Policy Uncertainty (EPU)** Index constructed by [Ghirelli et al. \(2019\)](#). The sample period spans from 2000M1 to 2025M12. Shaded areas represent 95% confidence intervals.

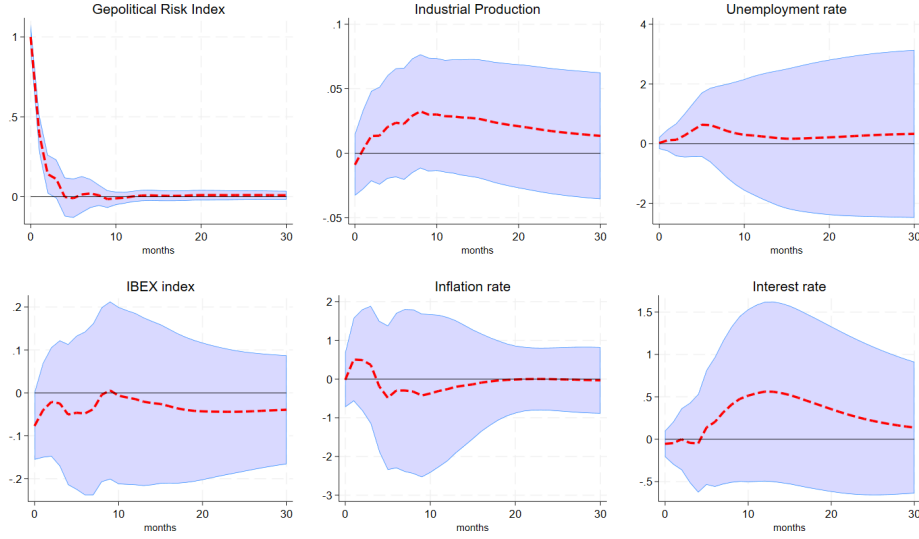


Figure A18: Impulse responses to a one standard deviation shock to the **Geopolitical Risk (GPR) Index** constructed by [Caldara and Iacoviello \(2022\)](#). The sample period spans from 2000M1 to 2025M12. Shaded areas represent 95% confidence intervals.

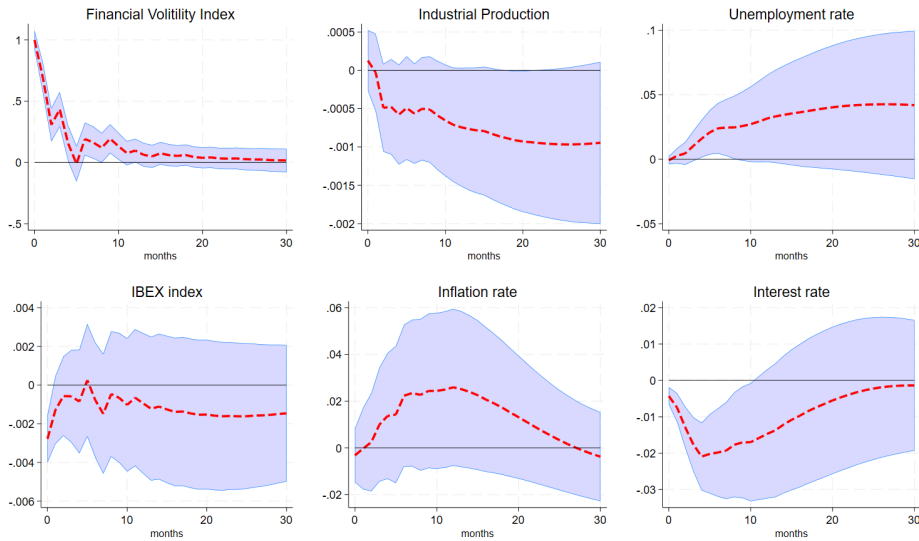


Figure A19: Impulse responses to a one standard deviation shock to the **Financial Volatility (VIBEX) Index**. The sample period spans from 2000M1 to 2025M12. Shaded areas represent 95% confidence intervals.

C.1 Principal Components Analysis (PCA)

This section reports the results from a principal components analysis of the ten topic-specific uncertainty indicators. Table [A3](#) presents the loadings of each topic on the first three principal components, highlighting the topics that contribute most strongly to each component. Table [A4](#) reports the eigenvalues and the proportion of total volatility explained by each component.

Variable	PC 1	PC 2	PC 3
Geopolitics	0.26	0.48	0.52
International trade	0.48	0.76	-0.26
Global demand	0.76	0.46	-0.23
Financial markets	0.81	-0.22	-0.04
Energy prices	0.54	-0.10	0.70
Inflation	0.84	0.10	0.31
Wages	0.60	-0.51	-0.09
Fiscal policy	0.63	-0.09	-0.31
Housing market	0.72	-0.47	-0.02
Confidence	0.89	0.01	-0.19

Table A3: Principal Component Loadings

Components	Eigenvalue	Proportion	Cumulative
Component 1	4.60	0.46	0.46
Component 2	1.60	0.16	0.62
Component 3	1.13	0.11	0.73
Component 4	0.91	0.09	0.82
Component 5	0.58	0.06	0.88
Component 6	0.39	0.04	0.92
Component 7	0.30	0.03	0.95
Component 8	0.27	0.03	0.98
Component 9	0.15	0.02	0.99
Component 10	0.08	0.01	1.00

Table A4: Proportion of Total Volatility Explained by Each Component